

Health systems strengthening II

Additional analyses and reflections

Global Health and
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Editorial note

This report was commissioned by Coefficient Giving (formerly Open Philanthropy) and produced by Rethink Priorities from February to March 2025. We revised the report for publication. Coefficient Giving does not necessarily endorse our conclusions, nor do our expert informants or the organizations with which they are affiliated.

This report serves as a follow-up to our previous investigation of health systems strengthening (HSS) ([Kudymowa et al., 2025](#)), which explored the case for HSS as a potential philanthropic cause area. The current report deepens that analysis by revisiting key uncertainties, developing new rough cost-effectiveness models, and incorporating additional expert interviews. We reviewed the scientific and gray literature and spoke with six experts, five of whom agreed to be named.

Throughout this report, we refer to concepts including “Coefficient Giving value” and Coefficient Giving’s cost-effectiveness “bar,” which are used within Coefficient Giving’s Global Health and Wellbeing portfolio to prioritize philanthropic causes. For a more detailed explanation of these concepts, we refer readers to [Oehlsen \(2024\)](#).

We tried to flag major sources of uncertainty in the report and are open to revising our views based on new information or further research.

Executive summary

Scope and context

This report is a follow-up to our previous shallow review of health systems strengthening (HSS) as a cause area ([Kudymowa et al., 2025](#)). It provides a deeper analysis of HSS's importance, neglectedness, and tractability (INT), as well as a rough geographical prioritization to identify countries where interventions may be most impactful. The focus is primarily on sub-Saharan Africa (SSA), but due to data limitations, we also incorporate broader analyses at the global or low- and middle-income countries (LMICs) level where necessary. Compared to previous analyses, this report places greater emphasis on higher-level systemic reforms rather than direct service provision.

Key takeaways

Importance

- In SSA, ~2 million deaths each year could be prevented with better healthcare access and quality (~8.6 million deaths in LMICs). The burden is especially concentrated in fragile and conflict-affected settings, including parts of the Sahel and Horn of Africa. These figures likely understate the true burden, as they exclude non-fatal outcomes, gaps in preventive care, and broader system failures beyond clinical services.
- Very rough estimates suggest the potential gains from HSS are very large. Across methods based on amenable mortality, excess burden relative to health spending, and convergence to better-performing peers, results point to substantial avoidable health losses, including excess DALY burdens of around 30–40% in the worst-performing countries and scenarios where system improvements appear highly cost-effective.

Neglectedness

- Experts consistently identified HSS as one of the most neglected areas in global health. Only around 7% of development assistance for health is directed toward health systems, with most funding instead going to disease-specific programs. Within the small share allocated to health systems, funding is heavily concentrated in service delivery infrastructure, while the health workforce remains underfunded relative to its importance.
- A key recent development is the withdrawal of USAID funding in several LMICs, which experts reported has triggered immediate disruptions to core health system functions, including stalled reforms, canceled programs, and deteriorating data and supply systems. USAID has historically played an outsized role not just in financing, but in providing embedded technical assistance for health financing, governance, and data systems, leaving a gap that many governments will struggle to replace.

Tractability

- Expert interviews suggest that recent USAID cuts have created a time-sensitive window for action, particularly in countries where donor funding supported core government capacity. At the same time, experts consistently cautioned against large-scale infrastructure, supply chain, and standalone digital investments, which they described as expensive, slow to implement, and difficult to align with government priorities.
- Our updated cost-effectiveness analysis reinforces this view. Earlier-modeled interventions such as supervision and mentorship and community scorecards remain

more promising than most newly explored supply chain, digital health, and incentive-based interventions, which generally fall well below Coefficient Giving's cost-effectiveness bar despite delivering operational improvements.

- Relative to our prior report, a key shift is toward viewing provider autonomy, embedded technical assistance, and strategic use of data as more tractable and promising than new infrastructure or tools. Health systems research, particularly embedded analytics and cross-country learning platforms, also appears unusually neglected and potentially high-leverage in the current funding environment.

Geographic prioritization and case studies

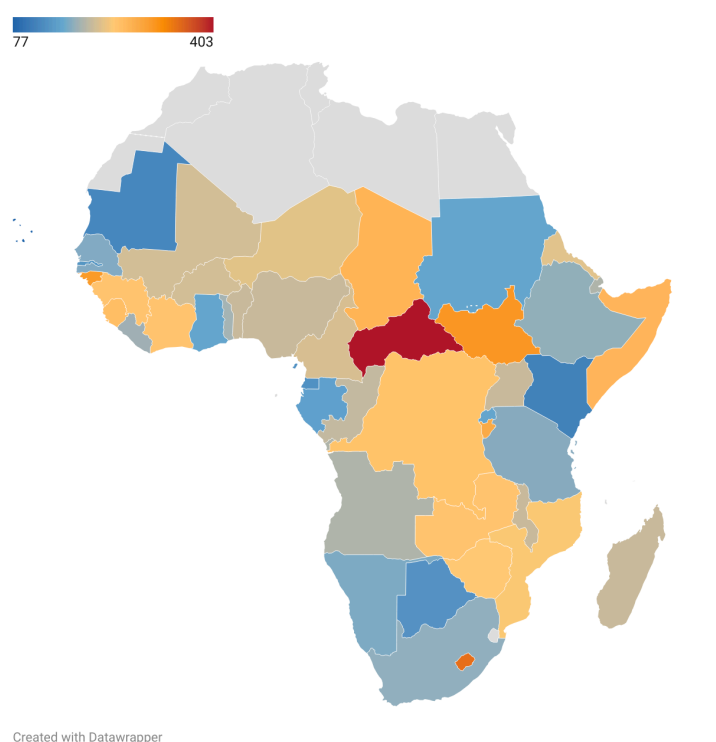
- A rough geographic prioritization identifies several high-burden, underfunded countries as potential candidates for HSS investment, though a full tractability assessment remains pending.
- Case studies of major funders and implementers illustrate two related strategies: vertical programs adding HSS components, and efforts to integrate vertical delivery into broader health systems. In both cases, reforms were shaped by political and financial constraints, unevenly implemented, and delivered mixed results, with limited evidence of sustained health impact.

Importance

Health burden attributable to poor health systems performance in LMICs

To understand the health burden associated with deficiencies in health systems across SSA, we draw on estimates of deaths amenable to healthcare from [Kruk et al. \(2018a\)](#). These estimates help quantify how many lives could be saved with better access to and quality of care (see [Figure 1](#)). To our knowledge, no other recent study provides comparable estimates of the health burden amenable to healthcare in LMICs. For a global perspective, [Figure A1](#) in the Appendix extends this analysis to 137 LMICs. A summary of [Kruk et al.'s \(2018a\)](#) methodology is provided in [Appendix B](#).¹

Figure 1: Deaths amenable to healthcare per 100,000 people across sub-Saharan Africa in 2018



Note. Calculated based on Table S5 in [Kruk et al.'s \(2018a; 2018b\)](#). Calculations [here](#). Interactive visualization [here](#).

Here are some key findings based on [Kruk et al.'s \(2018a\)](#) data:

- Deaths amenable to healthcare are highest in conflict-affected and fragile states, with the Central African Republic, South Sudan, Somalia, and the Democratic Republic of the Congo among the worst affected.
- A regional pattern emerges, with the highest burden in the Sahel region and Central Africa, particularly in Chad, Mali, Niger, and Guinea.
- The lowest amenable burden is seen in Southern Africa and parts of East Africa, with Mauritius, Seychelles, Botswana, and Rwanda having relatively fewer deaths associated with poor healthcare.
- In SSA, 2 million deaths are amenable to healthcare (8.6 million in LMICs).

¹ We initially considered including IHME's Healthcare Access and Quality (HAQ) Index as an additional proxy for health system performance across SSA ([Haakenstad et al., 2022](#)), but its methodology and estimates closely align with those of [Kruk et al. \(2018a\)](#). We therefore do not discuss it here.

We believe these figures are **useful for a rough cross-country comparison** of the relative weaknesses of health systems, offering insight into how well or poorly different countries deliver effective healthcare. However, **we do not recommend interpreting [Kruk et al.'s \(2018a\)](#) estimates of amenable deaths as a direct measure of the health burden caused by poor health systems in LMICs, or using them to compare this burden with other cause areas.** This would be misleading for several reasons:

- **Amenable deaths capture only part of the health system's impact.** They reflect deaths preventable through timely and effective clinical care but exclude broader systemic failures, such as weak infrastructure, underfunded prevention, and workforce shortages.
- **The line between healthcare and population-level interventions is often blurred.** The study excludes deaths preventable by interventions like vaccination or sanitation. But weak health systems often undermine these efforts too, so some deaths attributed to non-healthcare failures may still reflect deeper systemic weaknesses.
- **The analysis excludes non-fatal burdens.** It focuses on mortality and does not capture the effects of poor healthcare on disability or quality of life.
- **Cross-cause comparisons can be misleading.** Comparing weak health systems to specific diseases like malaria is problematic, since many disease deaths are caused or worsened by systemic failures. Without precise data to separate these effects,² such comparisons may distort what is driving mortality.

Overall, we believe these **figures underestimate the health burden associated with weak health systems**, though we have not attempted to quantify the extent of this underestimation.

Rough estimates of the benefits of health system reform

We used three approaches to roughly estimate the health burden that could be avoided by strengthening health systems in SSA. Each approach captures a different aspect of underperformance: [\(1\)](#) deaths amenable to high-quality healthcare, [\(2\)](#) excess mortality relative to what would be expected given current health spending levels, and [\(3\)](#) the burden that could be avoided if countries matched the outcomes and investments of better-performing peers.

[Table 1](#) below shows the estimated cG value of avoidable health burden under each approach for two low-performing countries that were included in all three analyses: Mali and Sierra Leone. Estimates range from \$120 billion to \$491 billion for Mali and from \$53 billion to \$199 billion for Sierra Leone. While the numbers vary widely, they all suggest that improving health systems could have large benefits. None of these estimates should be taken literally. Each relies on simplifying assumptions and partial data. Still, together they provide a rough sense of the scale of opportunity from health system reform.

Approach 1. cG value of improving healthcare across countries based on [Kruk et al. \(2018a\)](#)

To estimate the potential value of improving healthcare across SSA, we applied cG's valuation framework³ to estimates of deaths amenable to healthcare from [Kruk et al. \(2018a\)](#). This yielded a total **cG value of roughly \$6.5 trillion** for addressing amenable deaths in SSA, and

² While [Kruk et al. \(2018a\)](#) provide some disaggregation of amenable mortality across disease categories, we found it insufficient to clearly delineate between the burden of poor healthcare vs. diseases. For example, malaria is lumped under a category called "Other infectious diseases" (see Figure 2 in the study). The study also provides no regional disaggregation of the burden associated with different disease categories.

³ Calculated as: (Amenable number of deaths) x (32 DALYs per death) x (\$100,000 per DALY). See [Open Philanthropy \(2024\)](#) for cG's DALY valuation and [Favaloro and Berger \(2021\)](#) for cG's approach to translate DALYs into deaths.

around [\\$27 trillion](#) across all LMICs. In Nigeria alone, the estimated cG value is roughly \$1.2 trillion.

Table 1: cG value of avoidable health burden in Mali and Sierra Leone across three approaches

Country	[Approach 1] cG value of burden amenable to healthcare (billion \$)	[Approach 2] cG value of excess burden relative to health spending (billion \$)	[Approach 3] cG value of burden avoidable by imitating better performers (billion \$)
Mali	120	491	352
Sierra Leone	53	199	123

Approach 2. Excess DALY burden relative to health spending levels

Total excess DALYs, the additional burden these countries face due to underperformance relative to expected outcomes, can also be used to roughly quantify avoidable health losses due to inefficiencies in LMIC health systems. To calculate excess DALYs, we relied on [Mor \(2022\)](#), which analyzes the relationship between health financing levels (specifically, per capita pooled and out-of-pocket expenditures) and health outcomes across countries. The study estimates residuals from a log-log regression of DALY rates on these financing variables. The residuals represent the difference between a country's actual DALY rate and the rate predicted based on its health spending levels. We used positive residuals as a measure of underperformance and multiplied them by each country's population to estimate the total number of DALYs that would be avoided if the country performed at the expected level. This approach provides a rough quantification of the scale of health system inefficiencies in LMICs, highlighting where improvements in how health resources are spent (e.g., increasing pooled financing) could lead to significant health gains. This analysis is correlational.

[Table 2](#) shows the total estimated excess DALYs across the five most underperforming LMICs (i.e., those with the highest residuals), representing the additional burden these countries face due to inefficiencies in their health financing. These estimates reveal the high magnitude of avoidable health losses, **with some countries facing excess DALY burdens equivalent to $\geq 40\%$ of their total disease burden.**

Table 2: Estimated annual avoidable DALY burden in the five worst-performing LMICs

Country	Residual (excess DALYs per 100k people)	Total avoidable DALYs (millions)	Avoidable DALYs in % of total DALYs	cG value (billion \$)
Central African Republic	40,441	1.8	40%	184
Lesotho	36,518	0.8	43%	76
Chad	30,391	4.4	33%	443
Mali	27,314	4.9	31%	491
Sierra Leone	27,100	2.0	38%	199

Note. See [here](#) for our calculations. Residuals are taken from [Mor \(2022\)](#)

Approach 3. Adopting the investment and outcomes of a higher-performing country

In our third approach to this question, we [modeled](#) how maternal, infant, and under-5 mortality rates in poor performing countries (Niger, Mali, and Sierra Leone) would change if these countries adopted the health system performance of Ghana or Uganda, two countries with relatively strong outcomes (particularly maternal, neonatal, and under-5 mortality) for their level of development and healthcare spending. Using country-level mortality and expenditure data, we estimated the DALYs that could be averted under a counterfactual scenario where we assumed that each country achieved Ghana's or Uganda's health outcomes by increasing spending to the same levels. We found that the three poor-performers combined could avert as many as 9 million cG DALYs per year in this scenario.

In addition, we used these calculations to estimate the cost-effectiveness of transitioning to stronger health systems. Mali and Niger, which under-invest in healthcare, could potentially avert a DALY for \$19–\$55 if they achieved Uganda's outcomes at similar costs (around **1,800x to 5,300x cG cost-effectiveness**). While Sierra Leone spends as much as Uganda, quality improvements still appear to offer substantial potential.

Neglectedness: HSS funding streams and gaps

We use a three-pronged approach to investigate HSS funding streams. We first reviewed the scientific literature and existing data analyses for [aggregate HSS funding estimates](#) and trends. Next, we reviewed several [major organizations](#) in the HSS field and their key reports. Finally, we spoke with [several experts](#) to understand on-the-ground funding gaps, funding shifts over time, and areas of significant neglect. Moreover, we discuss the impact of recent [USAID funding cuts](#) on HSS programs, which several experts identified as particularly disruptive.

Please interpret our conclusions and takeaways with caution, as we are not highly confident in any of them given significant data limitations. One reason for this is that different organizations define HSS in substantially different ways. For example, some include pandemic preparedness while others do not. Additionally, some activities are labeled as HSS even though they are arguably vertical interventions focused on specific diseases or service areas. These inconsistencies make it difficult to accurately assess HSS funding and identify true gaps.

Expert perspectives on neglectedness

Experts consistently pointed to HSS as **one of the most neglected areas in global health**, particularly in contrast to vertical, disease-specific interventions. Jonah Goldberg, a PhD student at the Harvard [T.H. Chan School of Public Health](#), pointed to the imbalance between HSS and vertical program funding, noting that initiatives such as PEPFAR, Gavi, and the Global Fund often insulate specific disease areas from broader system deficiencies rather than reinforcing system-wide capacity. Similarly, Carrie Ngongo, a furloughed Senior Health Systems Strengthening Specialist at [RTI International](#), emphasized in our conversation that while family planning is likely the most neglected area of health funding,⁴ HSS is not far behind due to its political complexity and difficulty in quantifying impact.

Governments often struggle to independently finance core health system functions—a reality made more urgent by the recent exit of USAID from health systems support. According to Ngongo, countries heavily reliant on aid for HSS funding now face **immediate and severe gaps** after the withdrawal of USAID from the space. Goldberg similarly emphasized that while vertical interventions may receive some forms of backfill funding from other donors, the sudden loss of USAID and PEPFAR support for technical assistance and health information systems is likely to have major, long-term effects on overall health system performance.

Another potentially high-leverage area is technical assistance (TA). Goldberg further argued that among potential funding opportunities, TA for government health reforms is extremely neglected relative to its (high) leverage. Governments often lack the analytic and financial capacity to implement needed reforms without external support, yet USAID's cuts have left these gaps largely unfilled. Adam Salisbury, a Senior Researcher at GiveWell who has assessed such programs, noted that evaluating HSS and TA grants is significantly harder than for direct delivery interventions. Such programs are rarely well-documented, and no central list of HSS projects exists, making it difficult for donors to learn from past successes and failures.⁵ While the lack of systematic evaluation creates challenges, he emphasized that GiveWell still funds TA grants and does not believe their difficulty of measurement makes them inherently less attractive.

⁴ Similarly, Pierre Akilimali, speaking from the University of Kinshasa in the DRC, identified maternal and child health (MCH) as the most severely underfunded area of the country's health system, followed by chronic disease, immunization, and malaria. He highlighted the government's limited ability to scale MCH interventions and noted that the country's family planning programs are collapsing due to funding gaps, despite the DRC having the world's highest population growth rate.

⁵ In Salisbury's words, reaching confident assessments of these grants was difficult because of "big discrepancies between grantee and government M&E data, difficulties in assessing the counterfactual," and "actual grant activities taking on a different shape to what we predicted."

Aggregate HSS funding estimates and trends

HSS comprises only ~7% of development assistance for health funding, likely totaling more than \$4.5B annually, and declined over the past decade

The best resource of HSS funding streams we are aware of is the annual *Financing Global Health* report published by the Institute for Health Metrics and Evaluation (IHME), with the most recent data from 2023 ([IHME, 2024a](#)). The report and an interactive online visualization tool ([IHME, 2024b](#)) provide insights into development assistance for health (DAH) funding trends across different health focus areas and track disbursements from donors to intermediary organizations to recipient countries. Funding for HSS can be found under the HSS/SWAp (Sector-Wide Approach) category.⁶

According to [IHME \(2024a\)](#) estimates, **total DAH funding for HSS was ~\$4.5 billion in 2023. This marks a decline compared to the previous decade**, when annual HSS funding ranged between ~\$5.5-6.5 billion. The share of DAH funding allocated to HSS also declined: HSS accounted for 7% of DAH funding in 2023, down from double that share in 2019 (see [Figure A2](#)). However, given some limitations in the estimation procedure (see [Appendix C](#) for more detail), **we suspect that these figures might underestimate HSS funding prior to 2025, particularly for funding from outside of the US and building blocks⁷ other than the health workforce**. We are unsure about the extent of underestimation and have not prioritized further investigation, particularly given the significant shifts that have occurred this year as a result of [US and UK policy decisions](#).

The majority of HSS funding goes to infrastructure for service delivery, whereas estimates suggest the health workforce should receive the largest share to reach SDG health targets

We use the WHO ([2007](#), p. 3) framework to categorize HSS interventions. The framework divides a health system into six “building blocks”: (1) service delivery, (2) health workforce, (3) information, (4) medical products, vaccines & technologies, (5) financing, (6) leadership/governance. A major limitation of IHME data on HSS funding streams is that it does not allow for disaggregation by specific interventions or health system building blocks.

To our knowledge, the only recent study that has attempted such disaggregation is [Kraus et al. \(2020\)](#). The authors manually reviewed and categorized a subset of DAH activities⁸ in the Creditor Reporting System (CRS) database by the OECD for the year 2015 (the latest year for which data was available at the time of analysis) and categorized these activities into HSS building blocks based on project descriptions.

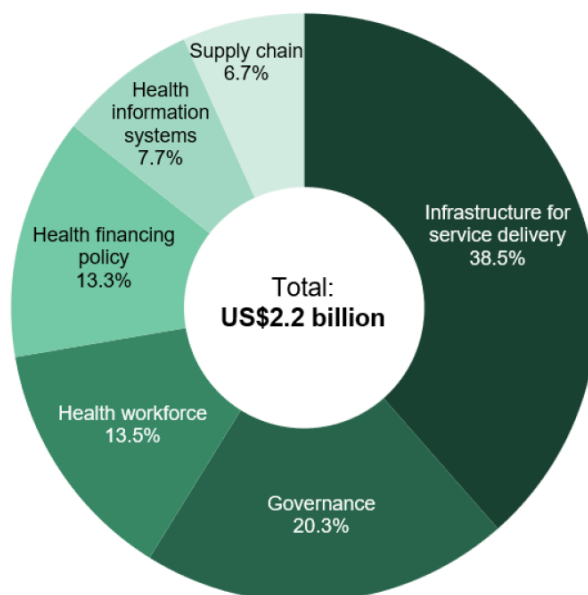
⁶ These terms are defined as follows: “DAH for health systems strengthening refers to funding intended to improve access, quality, or efficiency of health care, and at times emphasizes a specific health focus area or program, such as HIV/AIDS or family planning (RMNCH). Sector-wide approaches (SWAp) refer to funds that are pooled for broad, national goals such as monitoring and evaluating a health issue. In 2017, development assistance for pandemic preparedness was added to our tracking of health systems strengthening activities; this year, human resources for health has been added as a program area” ([IHME, 2019](#), p. 91).

⁷ According to the WHO ([2007](#), p. 3) framework to categorize HSS interventions, a health system consists of six “building blocks”: (1) service delivery, (2) health workforce, (3) information, (4) medical products, vaccines & technologies, (5) financing, (6) leadership/governance.

⁸ [Kraus et al. \(2020\)](#): “We reviewed 2,427 of 32,801 DAH activities in the Creditor Reporting System (CRS) database (80% of the total value of disbursements in 2015) and additional public information sources. Additional aid activities were identified through a keyword search.”

[Figure 2](#) below shows the 2015 distribution of HSS funding across health system building blocks as defined by the WHO ([2007](#), p. 3) framework. It shows that at that time, funding was distributed unevenly across building blocks, with infrastructure for service delivery receiving almost ~39% of all country-level DAH for HSS and supply chains receiving only ~7%. Unfortunately, we have not found more recent funding breakdowns, and it is possible that the distribution has changed over the past decade.

Figure 2: Distribution of system-wide HSS funding across major health system components in 2015



Constant US\$, 2015 prices. This figure does not include DAH for global HSS activities.

Note. Taken from Figure 4, [Kraus et al. \(2020\)](#)

While [Figure 2](#) offers a rough sense of how funding is distributed (or at least was until 2015), it does not indicate whether this allocation is appropriate or aligned with needs. To assess this, we need a clearer understanding of how funding should ideally be allocated. One suggestion comes from [Stenberg et al. \(2017a\)](#), which estimates the levels of investment needed in different HSS building blocks to achieve the Sustainable Development Goal (SDG) for health by 2030 (see [Appendix D](#) for a summary of the methodology). **They estimate that ~75% of resources need to be allocated to HSS, and only 25% to vertical programs⁹** (see [Figure 3](#)). This is in opposition to current DAH spending, with more than 90% allocated to vertical programs (see [Figure A2](#)), suggesting a significant imbalance in spending across the two categories and that HSS may be highly neglected.

If we focus on how funds should be allocated across different health system building blocks to reach health targets (see Section A in [Figure 3](#)), we also see that the distribution of resources may need adjustments (see [Appendix E](#) for a table overview of allocations). In 2015, infrastructure for service delivery appeared to be the largest building block, accounting for nearly 40% of HSS funding ([Kraus et al., 2020](#)), assuming allocations have remained relatively stable over the past decade. In contrast, [Stenberg et al. \(2017a\)](#) estimate that the health

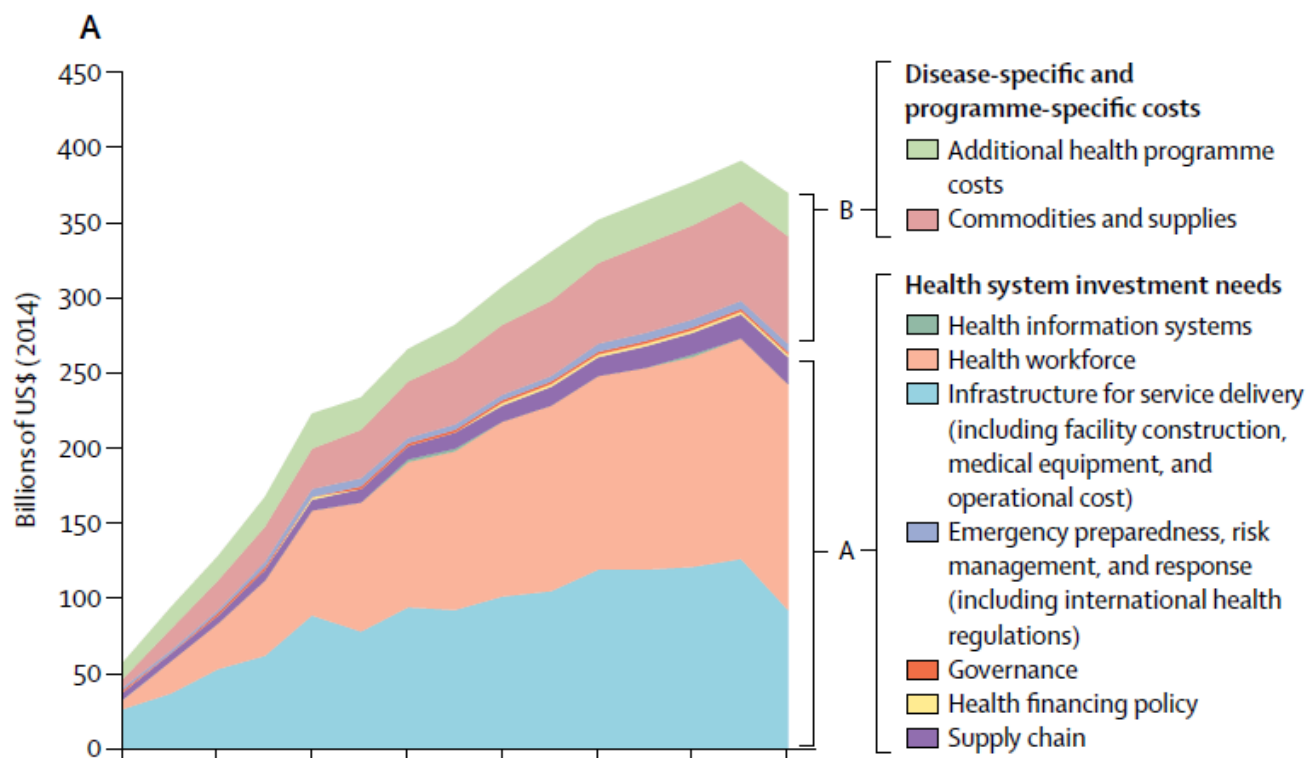
⁹ We distinguish between “horizontal” and “vertical” programs. In horizontal programming, healthcare is delivered via government-mandated and financed systems, commonly known as comprehensive primary healthcare or universal health coverage. In contrast, vertical delivery refers to the focused targeting of specific interventions, which are often not fully integrated into the broader health system and are frequently aligned with the priorities of the funding organizations ([Sakala et al., 2022](#)).

workforce requires the largest share (56%) of funding. Meanwhile, according to [Stenberg et al. \(2017a\)](#), governance, health financing, and health information systems each require less than 1% of HSS funding, but currently receive a significantly larger share (e.g., governance alone accounts for 20% of DAH funding for HSS). Notably, both estimates suggest a similar share of funding for supply chains. **These discrepancies suggest that to better align with health targets, reallocating resources, particularly toward the health workforce, may be worth considering.**

However, we are highly uncertain about this for several reasons:

- [Stenberg et al.'s \(2017a\)](#) estimates of resources needed to scale up health interventions likely skew toward areas with stronger evidence, potentially overlooking high-need areas with weaker data. Allocations for health workers and service delivery may be overestimated, while those for harder-to-evaluate building blocks like health information, governance, and financing could be underestimated. This pattern aligns with the evidence distribution noted by Witter et al. (2021, p. 112), who highlight disparities in research coverage across different health system components.
- [Stenberg et al.'s \(2017a\)](#) estimates allocate resources without considering cost-effectiveness as a criterion. Instead, their approach is based on modeling resource needs using existing treatment protocols and pre-defined intervention packages, rather than prioritizing investments based on comparative cost-effectiveness.
- Even if we accept [Stenberg et al.'s \(2017\)](#) allocation of health expenditures as “optimal” in total, it remains uncertain whether this allocation is optimal on the margin.
- We remain uncertain whether the allocation across building blocks estimated by [Kraus et al. \(2020\)](#) for 2015 still holds in 2025 or has shifted substantially.

Figure 3: Additional investments required in 67 low-income and middle-income countries to meet Sustainable Development Goal 3 (US\$ 2014 billion)



Note. Figure 2 from [Stenberg et al. \(2017a\)](#)

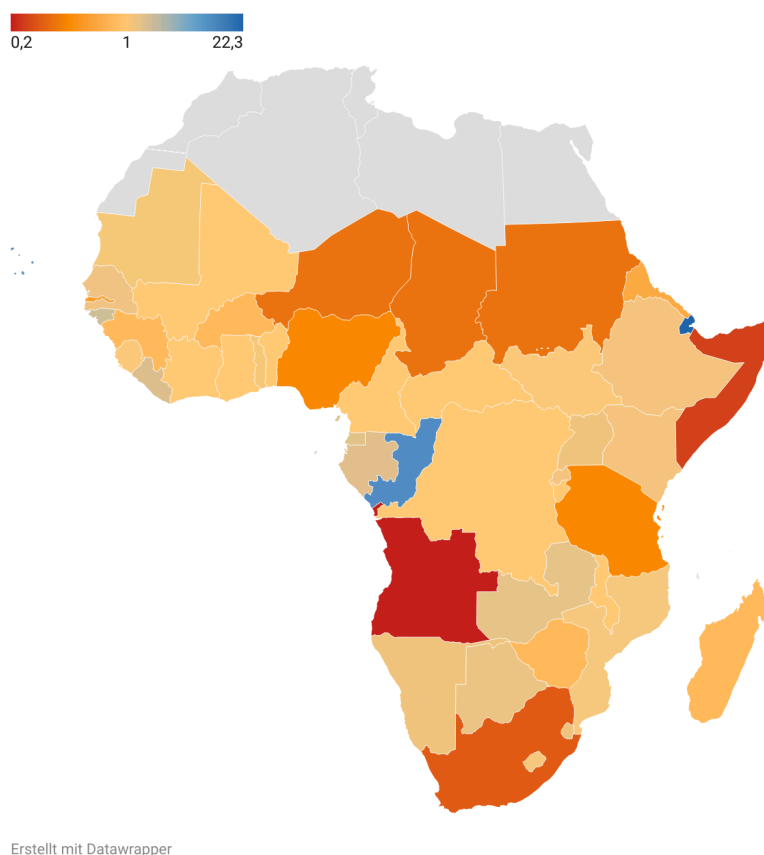
Sahel countries, Angola, Somalia, Tanzania, and South Africa receive the least HSS funding relative to their DALY burden

Understanding how HSS funding is distributed across countries provides insight into regional priorities and gaps. In the following, we compare HSS funding received relative to DALY burden across countries to investigate whether resources are allocated in proportion to health needs. We use HSS funding per DALY as a rough proxy for neglectedness, assuming that lower values indicate underinvestment relative to the disease burden.

[Figure 4](#) below shows the ratios between HSS/SWAp funding received and total all-cause DALY burden across countries in SSA (see [Datawrapper](#) for an interactive version with figures and country names). We can very roughly interpret this graph as **red countries being most neglected in terms of HSS funding and blue countries being the least neglected**, receiving relatively more funding per DALY. Here are some key takeaways:

- Several countries in Africa, particularly in the Sahel region (specifically Niger, Chad, and Sudan), as well as Angola, Somalia, Tanzania, and South Africa, emerge as the most neglected, with investments amounting to less than \$1 per DALY each. Notably, this overlaps partially with countries scoring poorly on the 2023 Fragile States Index ([Wikipedia, 2023a](#); including Chad, Sudan, Somalia, and South Sudan), where tractability is likely low.
- Cabo Verde, Congo, and Djibouti stand out as the least neglected countries, each receiving more than \$10 per DALY.

Figure 4: HSS/SWAp funding received/total DALY burden across sub-Saharan Africa in 2022



Note. Estimates on HSS/SWAp funding received and total all-cause DALYs per country are copied from [IHME's \(2022\)](#) Global Health Financing visualization tool for 2022. See [here](#) for our calculations.

Overview of key funders and organizations in the HSS space

We examined several major individual donors and organizations involved in HSS by reviewing their websites, strategic reports, and financial documents to identify their HSS activities and spending. Our goal was to understand roughly how much each organization allocates to HSS and which areas or activities they prioritize. This proved challenging as HSS funding is typically not clearly delineated within broader health budgets, and organizations vary in how they categorize and report these investments.¹⁰

Nonetheless, to the extent possible, we gathered key figures and examples to provide a broad sense of investment patterns across major global health funders, as summarized in [Table 3](#) below. We chose this non-exhaustive list of organizations based on their prominence as global health funders and IHME's (2022) estimated spending data on SWApS and HSS. **Our very rough best guess is that the actors in [Table 3](#) together spend ~\$4.7B on HSS per year** (roughly accounting for overlap between some of the individual funding estimates).¹¹ Given limited data, we are fairly uncertain about our estimated funding figures, and we are unsure whether we missed major other organizations.

Here are some takeaways/highlights:

- **The Gates Foundation's approach to health systems strengthening, emphasizing high-leverage, systemic reforms through government partnerships, advocacy, and technical assistance, offers a useful reference point.** Given cG's interest in supporting scalable and transformative improvements in health systems, examining Gates' strategy may help identify gaps, complementary opportunities, and areas where cG could add unique value.
- **The Global Fund is a major funder**, allocating \$2 billion annually to health and community systems, which represents approximately 30% of the organization's total annual funding ([Global Fund, 2024a](#)). We expect that a significant portion of this funding supports the health workforce,¹² but it also extends to other HSS building blocks.
- The World Bank is another major HSS player but does not seem to have a dedicated HSS unit. It funds large country-owned HSS projects. Thus, its HSS spending seems to be embedded in broader national health system investments.

There are several important caveats to these findings:

- **Discrepancies across estimates:** We noticed some very large discrepancies between IHME's HSS spending estimates and some organizations' self-reported HSS spending.

¹⁰ Many reports provide high-level figures without detailed breakdowns, making it difficult to understand which health system building blocks or interventions are prioritized. Additionally, some organizations integrate HSS into broader initiatives like universal health coverage or primary healthcare, further complicating our efforts to isolate direct HSS allocations.

¹¹ In our estimates, there is substantial overlap between Gates Foundation funding and organizations that receive funding from it. In 2020, ~20% of Gavi's total funding came from the Gates Foundation ([Wikipedia, 2025b](#)). Roughly ~7% of the Global Fund's funding comes from the Gate Foundation (see e.g., [Global Fund, 2025](#)). Moreover, ~9% of the WHO's funding comes from the Gates Foundation (see e.g., [WHO, 2022](#); [Carbonaro, 2023](#)). We expect that there is much less overlap for the other organizations (but we have not investigated this in detail). For example, we think that the funding figures from German, UK, and US official development cooperation in [Table 3](#) do not overlap with the other funding streams in the table, as we only used IHME's SWApS/HSS spending figures that are channeled via country-owned agencies (like USAID). Thus, we subtract the share of Gates funding from the other organizations' funding estimates to obtain a rough total funding figure of \$4.7B.

¹² This is because the Global Fund's annual spending on community health workers (\$300M/year) was the largest specific funding allocation we found mentioned in their reports ([Global Fund, 2024c](#)). See [Table 3](#).

For example, according to IHME, the Global Fund spent only \$113M on HSS in 2023, whereas the Global Fund's self-reported spending on HSS was \$1.8B ([Global Fund, 2024b](#)). In the same year, the Global Fund reports having spent \$150M on health information systems alone ([Global Fund, 2024c](#)). We are unsure about the source of this discrepancy, but one reason might be related to the IHME's limitations related to their keyword search, described in [Appendix C](#).

- **Double counting and data overlap:** Note that we cannot simply sum the funding figures above to estimate total HSS funding or rank the largest funders, as some organizations serve both as funders and implementers or serve as intermediaries between funders and national governments. This means the same funds may be reported at multiple stages of the financing chain, making it difficult to determine net investments in HSS.

Table 3: Overview of key funders involved in HSS

Organization	Annual HSS funding ^a	Key HSS focus areas and activities
Global Fund	~\$2B ^b for “Strengthening Health and Community Systems” ^c	HSS spending goes towards (Global Fund, 2024a): • Supply chains • Laboratories • Surveillance systems • Digital health information systems • Community health systems • Human resources for health We did not find a clear funding allocation breakdown, but found a few example figures for the 2024-2026 period (Global Fund, 2024d): • ~\$300M/year for community health workers^d • ~\$160M/year for laboratories • ~\$150M/year (from 2021-2023) for digital health information systems • ~99M/year (in 2023) for surveillance systems • ~9M/year for BIRCH project to support community health initiatives via technical assistance
World Bank	~\$1B/year ^e	They don’t seem to have a specific HSS unit, but several recent, large projects focus on HSS in various LMICs,^f e.g.: • ~\$1B for Indonesia for HSS project (IBRD; World Bank Group, 2023) • \$570M for Primary Healthcare Provision Strengthening Program in Nigeria (mostly IDA; World Bank Group, 2024a) • \$150M to strengthen health system in Burkina Faso (mostly IDA; World Bank Group, 2025) • \$100M to strengthen Resilience of Inclusive Health Systems for Everyone in Mali (IDA; World Bank Group, 2024b)
WHO	\$938M/year for universal health coverage ^g for 2024-2025 (WHO, 2023)	Focus areas within universal health coverage (WHO, 2025a): • Improved access to quality essential health services • Reduced number of people suffering financial hardships • Improved access to essential medicines, vaccines, diagnostics, and devices for primary health care The WHO has had a Special Programme on Primary Health Care since 2020 (WHO, 2024a). As of 2024, this program did, according to a recent WHO evaluation, not have a clear strategy or theory of change yet (WHO, 2025b, p. 2).
Gates Foundation	\$329M ^h	\$76M for primary health care (PHC) with following focus areas (Gates Foundation, 2023a): • Advancing foundational reforms for PHC • Implementing and scaling up accountability and quality measures • Testing innovative service delivery models • Scaling up impact through partnerships

		• Advocacy and communications Unclear which additional Gates funding falls into the HSS category to account for the ~\$250M difference between primary health care spending and the IHME estimate. From skimming recent grants, our impression is that many primary health care grants are focused on digital solutions (e.g., design health information platforms (Gates Foundation, 2024a); support effective use of data (Gates Foundation, 2024b); evaluate LLM for clinical decision-making (Gates Foundation, 2024c); optimize digital community health worker program (Gates Foundation, 2024d)).ⁱ
Gavi	>\$240M/year from 2021-2025 (Gavi, 2023 , p. 12)	HSS key strategic focus areas (Gavi, 2025): • Data • Supply chain • Leadership and management • Supply promotion Generally, our impression is that Gavi's HSS activities are heavily focused on improving immunization outcomes rather than broader health system improvements. See also here.
US development cooperation (mainly USAID + PEPFAR)	>\$200M ^j	Information on USAID funding and activities is hard to find as the website is defunct since US's foreign aid freeze, but here are some example activities from previous years: • Health Systems Strengthening Accelerator (2018); no longer active) • Strengthening Local Health Systems Activity in Uganda: Technical assistance to the Ugandan Ministry of Health to e.g., develop policies, strategies, and intervention packages to improve the performance of the health system with respect to health financing, health workforce, health information systems, and community health systems (\$40M over 5 years) • "Public Health Systems and Security" is one of PEPFAR's current strategic pillars (PEPFAR, 2022); see also here.
German development cooperation (mainly GIZ + KfW)	~\$169M ^k	HSS is a strategic priority for the German government, and it seems to be active across all HSS building blocks (The Federal Ministry of Health, 2020), but we found very little data on how the funding is allocated. Here are some example activities we came across: • BACKUP Health (GIZ, 2024): Program to assist countries in effectively using funds from global health initiatives such as Global Fund (technical assistance, capacity building, and policy development) • L4UHC leadership course: 1-year governance training to accelerate UHC progress among participating countries • Strengthening Health Care Worldwide (GIZ, 2023): mainly advisory services on how to improve health systems • Improve disease surveillance system in Liberia (GIZ, 2021): laboratories, training, improving disease surveillance
UK development cooperation (FCDO)	~\$68M ^l	The FCDO has been committed to HSS since at least its landmark position paper on HSS (FCDO, 2021), in which it affirmed its dedication to work on 13 different HSS focus areas. This position paper has been criticized (Drake & Baker, 2021) for being too broad and insufficiently prioritized, making it unclear which areas would receive the most focus and funding. Example project:

		Health Pooled Fund (HPF) in South Sudan (FDCO, 2020): delivering basic package of health and nutrition services, promoting community engagement, and supporting the stabilization of local health systems
Wellcome Trust	Likely in the low millions \$	The Wellcome Trust does not have a specific HSS unit, but they support(ed) several HSS interventions, such as: - Joint Health Systems Research initiative (Wellcome, 2016; discontinued in 2023): £4M/year (~\$5M) - Heightening Institutional Capacity for Government Use of Health Research Award (Wellcome, 2020; closed): £1M over 3 years (~\$1.3M); to evaluate methods for strengthening health ministries' capacities to make better use of health research - Strengthening health and disease modeling for public health decision-making in Africa (Wellcome, 2023): £3M over 3-5 years (~\$3.8M); for improving disease modeling and strengthening the modeling ecosystem in Africa

^a To estimate funding figures, we tried to mainly rely on each organization's own stated spendings. If we were unable to find this information, we relied on [IHME's \(2022\)](#) Financing Global Health, though we think these are likely underestimates.

^b "Between 2024-2026 the Global Fund is investing US\$6 billion to support countries to strengthen their health and community systems. This represents over one-third of all Global Fund investments in this period" ([Global Fund, 2024a](#)).

^c When the Global Fund writes about its HSS efforts, it sometimes refers to "Resilient and Sustainable Systems for Health" ([Global Fund, 2023](#)) and in other instances to "Strengthening Health and Community Systems" ([Global Fund, 2024a](#)). We are unsure what the difference between these terms is.

^d We were not able to quickly determine exactly how this is spent, but we have seen Last Mile Health mentioned several times ([Global Fund, 2024e](#); which we also briefly profiled in our [previous HSS report](#)).

^e A large share of the World Bank's spending is in the form of loans whereas other donors seem to focus on grants— these are not directly comparable. To make these more comparable, we break down the World Bank's spending into [IDA \(2025\)](#) (primarily focused on grants) and [IBRD](#) (focused on loans) and multiply IBRD funding by 10%, which is an approximation of the implied grant-equivalent for a World Bank loan.

^f The World Bank has different financing arms. The International Development Association ([IDA, 2025](#)) gives primarily grants and highly concessional loans. The International Bank for Reconstruction and Development ([IBRD](#)) provides grants.

^g See Table 2 (p. 3) in [WHO \(2025a\)](#): \$1,966M for universal health coverage in 2024-2025.

^h Estimated 2023 Gates Foundation spending on SWAp/HSS according to IHME global health financing estimates ([IHME, 2022](#)).

ⁱ See "Committed grants" filtered for "Delivery of Solutions to Improve Global Health" ([Gates Foundation, 2025](#)).

^j US' estimated 2023 spending on SWAp/HSS channeled via official US development cooperation according to IHME estimates ([IHME, 2022](#)). After skimming a list of recent USAID-terminated awards, we think that this is likely an underestimate ([USAID, 2025](#)). However, due to time and data constraints, we have deprioritized spending more time to get a better estimate based on the spreadsheet.

^k Germany's estimated 2023 spending on SWAp/HSS, channeled via official German development cooperation ([IHME, 2022](#)).

^l UK's estimated 2023 spending on SWAp/HSS channeled via official UK development cooperation according to IHME estimates ([IHME, 2022](#)).

Impact of recent USAID cuts in the HSS space

The abrupt withdrawal of USAID funding has created immediate and severe disruptions across multiple dimensions of HSS in LMICs. Historically, according to our interviewees, USAID was one of the most significant funders of HSS, particularly in technical assistance (TA), health management information systems (HMIS), and drug supply chains. Its departure has left governments and implementing organizations scrambling to maintain essential programs, with many key functions now at risk of deterioration.

USAID funding constituted a substantial share of HSS support (and indeed health-related support) in many countries. Pierre Akilimali, Associate Professor of Medicine and Public Health at the Kinshasa School of Medicine and the Kinshasa School of Public Health, estimated that 30-50% of total health system expenditures in the Democratic Republic of the Congo (DRC) came from USAID, particularly for medications, maternal and child health (MCH), family planning, and disease programs. In Senegal, RTI International had been implementing a USAID-funded program covering five regions and 5 million people, which was canceled overnight. These sudden funding gaps have jeopardized both direct service provision and the broader governance and technical support that underpinned these programs.

Experts emphasized that USAID's departure is particularly concerning because HSS programs are more difficult to replace than vertical health interventions. While malaria or HIV programs often have multiple funders with overlapping mandates, HSS initiatives, such as supply chain efficiency, health worker training, and financing reforms, tend to rely on a small number of technical donors. USAID played an outsized role in this space, making its withdrawal uniquely disruptive.

According to our interviews, **a major share of USAID's HSS investments went into technical assistance (TA) for government-led health financing and regulatory reforms.** This support played a critical role in shaping policies and ensuring accountability in how health funds were spent. Now, many of these efforts have stalled. In Kenya, USAID-funded technical staff within the Council of Governors had been leading county-level health financing reforms. These staff members, who were essential in pushing forward implementation, have now been furloughed or let go. In several other countries, USAID-funded advisors on social health insurance were leading efforts to design payment models and provider reimbursement rates. Without expert input, poorly designed models risk underfunding healthcare providers, reducing service availability, or even destabilizing the entire health system. The loss of oversight and accountability mechanisms also raises concerns about fund misallocation and corruption, particularly in environments where governance challenges are already significant.

HMIS and large-scale data collection efforts have also been deeply affected. Goldberg noted that virtually all Demographic and Health Survey (DHS) funding in SSA has come from USAID, while health management information systems (HMIS) have been supported by a combination of USAID and PEPFAR funding—much of the latter administered through the CDC rather than USAID. Both funding streams have faced significant recent cuts. While some governments may attempt to maintain these systems, the loss of donor funding threatens their functionality. Deterioration in data quality and availability is likely, as funding for software updates, training, and quality assurance has been eliminated. Without these systems, governments may struggle to track disease outbreaks, monitor drug stock levels, and allocate resources efficiently. Many countries relied on the US's technical and financial support to maintain early warning systems,

supply chain tracking, and health performance metrics. Without these inputs, according to Goldberg, decision-making will become less data-driven and more reactive, increasing inefficiencies. Indeed, one public health official noted in an interview that USAID's abrupt exit disrupted multiple priority programs, including support for DHIS2 interoperability and data-driven reforms.

Experts also noted that they expect essential medicine stockouts to increase dramatically, as supply chain financing mechanisms are affected. Akilimali noted that USAID was the primary funder for medication procurement in the DRC, particularly for malaria, HIV, and maternal health supplies. Similar challenges are expected across other LMICs that depend on USAID-funded drug procurement, warehousing, and distribution logistics. Beyond direct funding, he said, USAID also played a crucial technical role in improving procurement efficiency, reducing delays, and strengthening supply chain oversight. Without this expertise, countries may see higher costs, longer delays, and failures to distribute medicines where they are most needed.

Throughout our interviews, experts expressed deep concern about both the immediate service disruptions and the longer-term institutional weakening of health systems. Ngongo described the situation in stark terms, explaining that health programs covering millions of people were canceled with no transition plan in place. Akilimali warned that maternal and child health programs will suffer the most, as they were particularly reliant on donor support. Goldberg underscored the broader policy implications, emphasizing that poorly designed reforms, implemented without USAID-funded technical expertise,¹³ could have long-lasting negative consequences. While some governments may partially compensate for the loss of USAID funding, many will struggle.

The USAID cuts represent a seismic shift in global health funding, creating major gaps in financing, governance, and technical capacity. While some funders may step in to replace portions of this support, no single donor is expected to fully compensate for the loss. **For funders like Coefficient Giving, our experts stressed that these gaps present both urgent needs and strategic opportunities.**

Tractability

To assess which types of health systems interventions might be tractable for a philanthropic funder, we developed [rough cost-effectiveness models](#) and consulted [expert perspectives](#). [Table 4](#) summarizes our modeled estimates for selected interventions. An overview of key implementers and their estimated HSS spending is provided in [Appendix F](#).

Summary of modeled cost-effectiveness estimates

[Table 4](#) below summarizes the cost-effectiveness estimates from this round of modeling alongside select estimates from our previous report ([Kudymowa et al., 2025](#)). This phase focused on previously underexplored health system building blocks, including drug delivery, information systems, financing, and research. We deprioritized areas like workforce, service delivery, and governance, where we saw diminishing returns from further exploration.

¹³ Goldberg also noted that such support is often “provided through in-country contractors, who may be citizens of the countries in question, embedded in their government agencies, but whose salaries are paid by USAID. These folks would make much less if they were paid by their own governments, and in the absence of USAID, they might leave public service, taking their expertise with them.”

We modeled several supply chain interventions. **Structural reforms like Zambia's cross-docking system still stand out** (3,323x), though such examples are rare and context-specific. Most last-mile delivery efforts remain expensive and difficult to scale. Sierra Leone's vaccine program tripled coverage in remote areas but showed only modest returns in our model (398x). USAID's troubled logistics contract further underscored the risks of large-scale delivery investments. We see targeted, tech-enabled improvements as potentially valuable when embedded in broader reforms with clear accountability and cost structures.

We also prioritized health information systems. Many LMICs face fragmented, donor-driven systems that lack interoperability. While HR tools like iHRIS showed moderate cost-effectiveness (1,178x), digital decision-support tools such as mHealth apps performed less well (503x). Interviews reinforced the idea that health information systems (HIS) investments are most useful when they strengthen decision-making and feedback loops at the local level, not when they create additional burdens or fragmentation.

In financing, our model of results-based financing (RBF) yielded the highest estimate this round (1,305x), though still below cG's bar. This calls into question the stronger estimate for community scorecards (CSCs) from the previous report. Experts suggested that the main value of RBF may lie in the autonomy it gives local providers, rather than the financial incentives themselves. Several pointed to an acute gap left by the collapse of USAID's technical assistance ecosystem, particularly in public financial management, procurement, and budget execution.

We did not model research-focused interventions, but multiple experts highlighted the collapse of traditional data systems (e.g., DHS surveys) and the need for policy-embedded research. Areas of interest include implementation-linked evaluations, cross-country learning platforms, and political economy work on policymaker behavior. Strategic support for such research may be both tractable and urgent in the current funding climate.

As shown in [Table 4](#), earlier models of CHW programs (997x), IMCI (2,659x), and supervision and mentorship (8,546x) continue to rank among the most promising interventions, outperforming most of the interventions modeled in this recent round.

Table 4: Summary of rough cost-effectiveness estimates for selected interventions

Intervention	Assessed cost-effectiveness	Health system building block
Supervision and Mentorship for IMCI Providers	8546 x	Leadership and governance
Drug delivery reform (Zambian cross-docking)	3300 x	Supply chains
Social Accountability / Community Scorecards (CSCs)	3123 x	Leadership and governance
Integrated Management of Childhood Illness (IMCI)	2659 x	Service delivery
Results-based financing for health centers	1305 x	Financing
HR management system	1285 x	Information systems
Community Health Workers (CHWs) and Service Extensions	997 x	Health workforce
mHealth to guide CHWs	503 x	Information systems
Continuous Quality Improvement (CQI)	470 x	Service delivery
Sierra Leone last-mile delivery	398 x	Supply chains
eVIN in India	248 x	Supply chains

Note. All cost-effectiveness is expressed in cG units. Darker grey indicates models built for this report (calculations [here](#)); light grey shows models from our previous report ([Kudymowa et al., 2025](#)).

Expert perspectives on tractability

Insights from our expert interviews suggested that certain HSS investments may be highly impactful but require careful selection to ensure feasibility. One of the clearest takeaways from expert discussions is that the **recent USAID cuts have created a window of opportunity for targeted interventions, particularly in countries where a substantial share of health system expenditures was donor-funded**. We discuss this issue in more depth [here](#).

Another key theme is that not all **HSS investments are equally viable**. Ngongo cautioned that supply chain improvements can be particularly costly, and in isolation their impact can be difficult to quantify. She suggested focusing on evidence generation and evidence-driven prioritization. Similarly, Akilimali noted infrastructure gaps but saw provider incentives, data quality, and facility management as more feasible targets than major capital investments.

Somil Nagpal, Lead Health Specialist at the World Bank's East Asia & Pacific (EAP) region, and Goldberg both mentioned **technical assistance (TA) as a cost-effective way to strengthen health systems**. Goldberg emphasized that some countries have lost key health financing, governance, and procurement advisors due to USAID cuts, creating a pressing need for policy expertise and reform guidance, and mentioned the specific case of rate-setting for social health insurance schemes. Both Nagpal and Goldberg pointed to **peer-learning networks as an efficient way to provide TA at scale**, citing the Joint Learning Network for UHC, which they mentioned as a **high-leverage opportunity, reaching 40 countries at a relatively low cost**.

Health product supply chains

Access to essential medicines remains a critical constraint in many LMIC health systems. Several experts for this report identified drug stockouts and weak pharmaceutical logistics as both urgent and tractable problems.¹⁴ Goldberg noted that “**stockouts of drugs and supplies are a meaningful constraint in the public sector in a lot of countries,**” and that **technical assistance to address this is likely to be highly cost-effective**. Nagpal also emphasized that even where supply chains exist, they often fail due to poor planning and weak incentives.¹⁵ Akilimali echoed this, warning that the loss of USAID support could severely disrupt drug procurement and distribution in the DRC.

In our previous report on HSS ([Kudymowa et al., 2025](#)), we explored health supply chain interventions primarily through a review of the scientific literature, focusing mainly on two key systematic reviews of various health supply chain interventions.¹⁶ The evidence suggests that supply chain interventions can lead to significant cost savings and increased drug availability; however, direct evidence on improved health outcomes is lacking. While some reforms (e.g., Zambia’s “cross-docking” reform, see [Kudymowa et al., 2025](#), p. 40) and interventions have achieved impressive results, generalizable conclusions are difficult to draw, as effectiveness appears highly context-dependent. Mixed results and frequent experimentation with different supply chain approaches/structures¹⁷ highlight the complexity of reform efforts.

In our further research on health product supply chains, included in this report, we have tried to develop a deeper understanding of the field by reviewing how supply chains work, the main supply chain failures, and proposed solutions (e.g., [Yadav, 2015](#); [Silverman et al., 2019](#)). We also asked experts about both the potential and downsides of interventions to improve health product supply chains. Nagpal described a successful results-based financing intervention in Laos, where districts were rewarded for reducing stockouts of essential medicines (we further discuss results-based financing [below](#)). In the Laos case, managers began actively tracking inventory and coordinating across districts to reallocate stock as needed, and according to Nagpal, more than 80% of districts stopped reporting stockouts. Akilimali and Ngongo, however, said that drug supply reforms must be paired with better information systems and strong oversight, noting the risk of corruption or inefficiencies without clear financial and procurement accountability. We excluded two categories of interventions from our analysis, pooled procurement¹⁸ and global market-access initiatives.¹⁹

See [Appendix G](#) for an overview of how health supply chains are organized in LMICs, who pays for drugs, and what factors drive the costs of health products. See also [Appendix H](#) for an

¹⁴ Salisbury highlighted several recurring supply chain bottlenecks, including delayed procurement planning, weak quality assurance in low-resource settings, and unreliable demand forecasting, particularly when bottom-up estimates are inflated due to incentive misalignment.

¹⁵ Nagpal also highlighted the potential of emerging technologies, such as drone delivery and AI-assisted stock tracking, but warned that basic logistics (like electricity and connectivity) must be addressed first.

¹⁶ [Nunan & Duke \(2011\)](#), and [Seidman & Atun \(2017\)](#).

¹⁷ In our previous HSS report, we illustrated this with an example from Uganda where even after decades of experimentation with different supply chain structures, no clearly superior system emerged ([Kudymowa et al., 2025](#), p. 35).

¹⁸ This was excluded as it had already been covered in a previous Rethink Priorities report on government procurement ([Van Schoubroeck et al., 2024](#)).

¹⁹ We excluded interventions that primarily aim to influence global health markets rather than directly strengthen health systems. This includes programs such as the Affordable Medicines Facility-malaria (AMFm), the Health Impact Fund, the Global Financing Facility, and the Medicines Patent Pool. While these initiatives play an important role in expanding access to health products, they largely work in parallel to national health systems.

overview of the main supply chain failures in LMICs. Additionally, we reviewed the [Gates Foundation's \(2025\)](#) committed grants related to supply chain strengthening to better understand the types of interventions that are currently prioritized in the field.²⁰

While the space of supply chain strengthening is vast, we observed similar themes across sources, which can be roughly categorized into three broad areas: **(1) structural & governance reforms, (2) operational & logistics efficiency, and (3) technology & data systems.** We summarize our findings for these three intervention categories in [Table 5](#) below. Many (if not most) of the interventions/organizations in the table are also funded by the Gates Foundation.

Structural & governance reforms

These reforms refer to high-level systemic changes to restructure the supply chain system in a major way. This category includes reforms such as reducing the number of tiers in the supply chain, decentralizing procurement, or introducing privatization. We think these are likely the highest-leverage opportunities, but they are also extremely context-specific, making it difficult to predict what will work well in different settings. This category has the least rigorous evidence base, as reforms tend to be long and complex.

We were initially optimistic about high-level structural supply chain reforms, as a previous rough model suggested that a recent national drug delivery reform in Zambia surpasses cG's cost-effectiveness bar at ~3,300x ([Kudymowa et al., 2025](#), p. 37). However, **we have struggled to find other reforms with similarly strong evidence of success or promise as the [Zambian reform](#).** Here are some examples of reforms that initially looked promising to us, but then much less so upon closer look:

- The Delhi rational drug use reform, launched in 1994, showed some promise in reducing drug costs and improving drug access (increased drug availability from 40-70% to >90%) through interventions like pooled procurement, standard treatment guidelines, and prescriber training ([Chaudhury et al., 2005](#)). However, a cost-benefit analysis suggests that the program was not cost-effective, and its benefits were limited to only 18% of the population.²¹ Unfortunately, the source of this claim doesn't provide any specific threshold for cost-effectiveness, and we have not been able to access the original source. Additionally, the reform faced implementation challenges, sustainability concerns, and limited scalability ([Jana et al., 2006](#)).
- Several decades of experimentation with different variants of “push” and “pull” systems for ordering drugs in Uganda did not result in a consistently effective or scalable solution, as both approaches faced chronic supply chain inefficiencies, stockouts, and implementation challenges ([Tumwine et al., 2010](#); [Bukuluki et al., 2013](#)).

TA for government-led reforms is one of the main ways to engage in structural and governance strengthening. Organizations like [CHAI](#), [IntraHealth \(2023\)](#), [Abt Global](#), and [MSH](#) frequently support Ministries of Health with reforms in health financing, regulation, and governance. Experts we interviewed viewed this as potentially highly cost-effective, particularly in light of the urgent gaps created by USAID's withdrawal from many TA efforts. However, while TA seems promising, we found few clearly documented success stories beyond the [Zambian reforms](#), and Ngongo cautioned that this type of support can be expensive to do well.

²⁰ We focused on the Gates Foundation as it is one of the largest funders in global health, and its publicly available grant database provides a comprehensive overview of recent investments. Given the Foundation's broad engagement in areas relevant to our analysis, we consider it a useful reference point.

²¹ “A cost benefit analysis of spending such a huge sum (Rs. 1.5 crore) and the extent of the benefit to the people of Delhi would show that the programme was not cost effective” ([Jana et al., 2006](#)).

Operational & logistics efficiency

This intervention category centers on the movement of health products, ensuring they reach the right place at the right time efficiently and cost-effectively. This includes innovations in last-mile delivery, improved distribution models, and better warehousing and inventory practices. While these interventions have immediate, measurable benefits, they often depend on complementary governance or technological improvements to be sustainable. We focused on two interventions in this category in more detail, [last mile-delivery interventions](#), and a very large-scale [USAID supply chain initiative](#).

Last-mile delivery interventions

Our impression is that last-mile delivery interventions are currently especially popular in health supply chain strengthening (judging by recent media coverage and academic research²²), possibly because last-mile delivery costs can be very high²³ and have potential to be reduced by technological innovations. We have seen several recent Gates Foundation grants²⁴ towards such interventions, and there are various prominent examples of organizations currently working on this (e.g., [VillageReach](#), [Zipline](#), [Riders for Health](#), and [LifeBank](#)).

We reviewed Sierra Leone's last-mile delivery intervention in more detail (see [Appendix I](#)), which aimed to increase Covid-19 vaccination rates in remote communities. We focused on this intervention because it has been rigorously evaluated in an RCT, and there is ample readily available data to estimate its cost-effectiveness. Although this intervention had **very impressive effects, with the vaccination rate being tripled within 48-72 hours**, a [rough model](#) suggests that it is far from meeting cG's cost-effectiveness bar at 398x.²⁵

We modified our model in several ways, such as assuming cheaper drone delivery instead of manned delivery and bundling the COVID-19 vaccine with another vaccine, to see whether these assumptions could make the intervention sufficiently cost-effective. However, even with these adjustments, the intervention still did not come close to meeting cG's cost-effectiveness bar, as operational costs remain fairly high. On the other hand, many last-mile delivery interventions are implemented by for-profit companies (e.g., [Zipline](#)), which means that these may not need funding for their recurrent operational expenses once they are established. Instead, cG could consider **providing seed funding during the start-up phase** to help get such organizations off the ground. This could potentially look similar to the Gates Foundation's ([Empower Africa, 2023](#)) recent seed funding of \$50k each for 29 health supply chain start-ups in Africa. One such example is Pharmarun ([Michael, 2023](#)) in Nigeria, which is described as "Uber in the pharma sector".

USAID's Global Health Supply Chain Procurement and Supply Management project (GHSC-PSM):

In 2015, the USAID awarded a \$9.5B contract to a consortium led by Chemonics International to strengthen supply chains in LMICs, the largest single contract in USAID history. We discuss the GHSC-PSM here because it serves as a real-world example of the complexities and challenges involved in designing and operating a large-scale supply chain strengthening program, even with substantial funding. The program focused on improving the

²² E.g., [Serwer \(2024\)](#), [Dludla \(2024\)](#), and [Meriggi et al. \(2024\)](#).

²³ "The last mile accounts for about 40% of total logistic costs globally" ([Saguna et al., 2021](#)).

²⁴ E.g., Zipline ([Gates Foundation, 2015](#)), Last Mile Health ([Gates Foundation, 2017](#)), Kaizen Institute Consulting Group ([Gates Foundation, 2024e](#)), Dev-Afrique Development Advisors ([Gates Foundation, 2024f](#)), [Project Last Mile](#), and Financing Alliance for Health ([Gates Foundation, 2023b](#)). Their type of support varies across grants (e.g., research, skill-building, and direct operational funding).

²⁵ This is true if recurrent operational costs are used as the cost metric. However, since these reflect direct service delivery, they may not align with cG's focus on leveraged, system-level impact.

procurement, distribution, and management of health commodities to ensure that essential health products reach people in LMICs, particularly those in remote areas. It operated in over 60 countries and was focused on various activities, such as technical assistance, capacity building, optimization of supply chain management, last-mile delivery strengthening, and implementation of digital tools. The project aimed to improve supply chains “to the point that they could be managed by the countries themselves” ([Igoe et al., 2023](#)).

However, the project failed drastically due to a long series of operational missteps and mismanagement, both by USAID and Chemonics International ([Igoe et al., 2023](#)). In its first year, the program saw a catastrophic drop in performance, with the percentage of shipments delivered “on time and in full” dropping from 67% to 7%. **This resulted in widespread stockouts of essential health supplies**, including HIV drugs and mosquito nets. Chemonics was criticized for poor communication, lack of coordination, data manipulation, and alleged fraud. While USAID recognized issues early on, due to the scale and the design of the contract, they had limited power to make meaningful changes and found themselves in a situation where the program was seen as “too big to fail”, but there were no clear ways in which they could hold Chemonics accountable or reallocate responsibilities. Even though the project was initially designed with a five-year timeline in mind, it was extended multiple times until at least 2024.

Despite attempts to recover, the project ultimately failed to meet its goals and became a cautionary tale. This forced USAID to launch a new, redesigned initiative, NextGen ([Ainsworth, 2022](#)), to overhaul the entire approach and replace GHSC-PSM. NextGen is an even larger, 10-year project at \$17B. Contracts have been awarded in 2024, though we suspect that it was very likely affected by USAID’s funding cuts in 2025. According to [Pisa and McCurdy \(2019\)](#), the GHSC-PSM’s “struggles illustrate the difficulty of getting global health supply chains ‘right’”.

Technology & data systems for drug delivery

This category refers to a wide range of digital solutions to improve forecasting, tracking, and overall supply chain efficiency. There are a large number of interventions in this space, including dozens of recent start-ups in SSA.²⁶ While many of these tools only had modest success in the past,²⁷ some show stronger potential (e.g., eVIN ([UNDP, 2024](#)), [Maisha Meds](#), and [mPharma](#)).

India’s electronic vaccine intelligence network (eVIN)

eVIN is a digital vaccine management system implemented in India to improve the efficiency and reliability of the immunization supply chain. It stood out to us as a widely recognized success story, reducing stockouts, enabling real-time inventory tracking, and operating effectively at national scale. Unlike many digital health tools that remain pilots, eVIN has been successfully scaled and institutionalized (e.g., [Yadav, 2024](#)). The system was developed and implemented by India’s Ministry of Health and Family Welfare, the United Nations

²⁶ The Gates Foundation recently announced ([Empower Africa, 2023](#)) that it will fund 29 healthcare supply chain start-ups in Africa, many (if not all) of which focus on digital tools.

²⁷ [Yadav \(2024\)](#): “However, despite the substantial potential for impact indicated by the successful examples presented in the preceding sections, the actual realized effects of digital technology on health product supply chains have generally been modest to date. Only a handful of noteworthy national-scale implementations have delivered their promised impact. Many countries have been slow to embrace digital solutions for the health product supply chain, as they feel ill-prepared to navigate their complexities and intricacies. As a result, there are currently only a few LMIC health supply chains that offer systematic evidence of digital technology adoption leading to long-term improvements in product availability, efficiency, affordability, or the expedited adoption of new health technologies.”

Development Programme, and Gavi. It digitizes vaccine stock tracking to enable real-time monitoring of vaccine availability, helping to reduce stockouts and wastage while improving supply chain transparency. Moreover, eVIN incorporates cold chain temperature monitoring, which helps maintain vaccine quality and prevent spoilage. In 2014, it was piloted in 12 Indian states and was later rolled out nationwide, covering 32 states and 23,500 cold chain points.

eVIN has been assessed through several before-after evaluations (e.g., [Gurnani et al., 2021](#); [Gurnani et al., 2020](#); [UNDP, 2018](#)), mainly from an economic perspective. Here are some key findings, but these are correlational rather than causal:

- Facilities experiencing vaccine stockouts decreased by 30.4%.
- Vaccine wastage was reduced by ~90 million doses.
- The economic return on investment was projected at \$2.93 per \$1 spent due to reduced vaccine wastage and improved stock management.
- ~400k children were prevented from missing immunization due to reduced stockouts.
- According to [Yadav \(2024\)](#), “visibility through eVIN was a contributor to the fast introduction of rotavirus vaccine in India in 2016”.

We built a [rough model](#) to determine if eVIN could meet cG’s bar. We consider both economic benefits (from savings due to reduced wastage and improved stock management) and health benefits (from additional children vaccinated due to reduced stockouts). We did not model any potential speedup benefits (e.g., in the case of the rotavirus vaccine). Our model suggests that while digital systems may reduce waste and inefficiencies, their indirect impact on mortality and health outcomes remains comparatively modest. **The low cost-effectiveness score of the eVIN program (~248x) raises doubts about the standalone value of digital health logistics interventions. Given these results, we now see strong reasons to deprioritize supply chain tracking tools as independent interventions, unless they are directly linked to major efficiency improvements in procurement, distribution, and frontline service delivery.**

Table 5: Overview of health product supply chain interventions

Intervention category	Example interventions	Example organizations	Successful case studies	Important considerations (+ = pro, - = contra, ! = other consideration)
Structural & governance reforms	- Reducing the number of supply chain tiers - Decentralizing procurement to lower administrative levels - Privatization or public-private partnerships - Strengthen regulatory frameworks	Africa Resource Centre: technical assistance and advisory services to health ministries PFSCM: technical assistance, but also procurement agent and supply chain services provider Management Sciences for Health (2024): provide countries technical assistance for supply chains for next 10 years (likely shut down due to USAID funding cuts)	- Zambia: ‘Cross-docking’ reform reduced stockouts from ~48% to ~13% (Vledder et al., 2019) - India: Comprehensive reform increased drug availability from 40-70% to >90% (Chaudhury et al., 2005) - Uganda: ‘Pull system’ for ordering drugs at hospital -> average % out-of-stock days reduced from ~15% to ~3% (Tumwine et al., 2010)	+ Potentially high leverage and long-term impact - Highly context-specific, hard to generalize findings across countries - Requires deep engagement with governments - Slow-moving reforms and political complexity ! While there is theoretical evidence on what needs to be done, there is very little written empirical evidence on what works well in practice
Operational & logistics efficiency	- Direct-to-clinic delivery models - Outsourcing logistics to private distributors - Using drones and other innovative transport solutions - Cold chain optimization for temperature-sensitive products (e.g., vaccines)	VillageReach: strengthens last-mile delivery through dedicated logistics systems (DLS) Zipline: uses drones to deliver medicines in remote areas Riders for Health: provides motorcycles for health workers LifeBank: multimodal distribution network (e.g., drones, trucks) for rapid supply of medical products to hospitals	- Simple last-mile vaccine intervention in Sierra Leone tripled vaccination rates within 48-72h (Meriggi et al., 2024) - Zipline: vaccine stockouts 60% shorter; patient 42% less likely to miss vaccination opportunity (Zipline, 2025) - Zimbabwe’s DTTU (Delivery Team Topping Up) system reduced stockouts of contraceptives by shifting to direct-to-clinic delivery from 20% to 2% (USAID, 2008)	+ Often high-impact and improvements directly measurable - Many logistics solutions require sustained funding - Operational costs of last-mile delivery interventions can be high ! cG may potentially be more interested in scalable tech-driven solutions (e.g., Zipline’s drones) rather than direct service delivery (e.g., fleet of vehicles)

Technology & data systems	• Electronic logistics management information systems (eLMIS) • AI-driven forecasting tools to predict demand • Digital pharmacy inventory management	• E-VIN (UNDP, 2023): vaccine supply tracking system • mPharma: stock management support for pharmacies • Maisha Meds: digitizes pharmacy operations for better stock taking • Zenysis Technologies: uses AI to integrate supply chain data	• E-VIN digitized vaccine stock and consumption data, reducing stockouts by ~30% and preventing waste of ~90M vaccine doses (UNDP, 2018)	+ Highly scalable + Can be relatively cheap²⁸ - Requires strong health system integration and long-term adoption - Many tools had only modest success in the past - Unclear sustainability beyond pilot phase (some tech interventions work in small-scale pilots, but struggle in national scale adoption)
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²⁸ The Gates Foundation committed grants of \$50k each to 29 African healthcare supply chain start-ups.

Health information

A well-functioning health information system (HIS) is essential for improving health system performance, enabling better decision-making, resource allocation, and service delivery. In theory, a strong HIS should ensure timely, accurate, and complete data collection, allow seamless interoperability between digital health tools, and support disease surveillance, supply chain management, policymaking, and financing decisions. However, in practice, many LMICs struggle with fragmented, underfunded, or outdated health information systems.

Our prior research identified health information systems as a critical but underfunded building block ([Kudymowa et al., 2025](#)). Without reliable data, governments and implementers cannot efficiently allocate resources, identify gaps in service delivery, or respond to emerging health crises. Despite this, HIS initiatives frequently suffer from poor interoperability and inconsistent adoption, raising concerns about their long-term impact. Experts stressed that strengthening data systems would enable better decision-making and resource allocation, but also noted that these systems require long-term investment rather than one-off grants. In our further research into health information systems, we have found that many promising systems have proliferated across a range of use cases in the space, leading to serious issues with fragmentation and unsustainability.

One commonly cited bottleneck is **tool fragmentation and lack of interoperability**. Several experts highlighted that most LMICs are operating with dozens of uncoordinated digital platforms, often built for donor priorities rather than system-wide utility. For example, malaria programs may use bespoke monitoring tools, while maternal health initiatives rely on standalone mobile apps, creating parallel, siloed data systems that fail to communicate with each other. The lack of interoperability between donor-funded tools and national health management information systems has led to inefficiencies, data duplication, and gaps in national reporting ([Gambo et al., 2011](#), p. 185).

Our impression is that investing in systems that reduce duplication, improve data-sharing, and track drug availability could yield better long-term benefits than sponsoring a nascent tool. Ngongo, Akilimali, and one other expert all emphasized the importance of routine health data systems like DHIS2, which are increasingly country-owned but still reliant on donor funding for staffing, training, and infrastructure. However, adoption varies significantly: some countries have integrated DHIS2 effectively into their national health infrastructure, while others still rely on fragmented, donor-specific data collection systems ([Serge et al., 2024](#), p. 7).

Furthermore, data collection is often incomplete, outdated, or underutilized in research, system monitoring, and policy evaluation ([Hung et al., 2020](#)). Some places still rely on paper-based records or outdated software, leading to delays in reporting, difficulties in tracking stock levels, and missed opportunities for data-driven decision-making ([Msiska et al., 2017](#), p. 247). Another key insight from Nagpal is that better data alone is not enough. Many LMICs already collect large quantities of health data but lack the analytical capacity or institutional culture to use it effectively. He argued that **HIS investments should go beyond data collection to focus on building feedback loops and real-time decision support**— systems that guide district managers, clinic administrators, and national policymakers toward better performance. Ngongo emphasized that without proper training and incentives, many providers are unlikely to value data or see it used meaningfully in decision-making. As a result, in some contexts the quality of DHIS2 data can be poor and unreliable. There may be opportunities to support

scalable approaches that enhance data quality, strengthen feedback loops, and rebuild trust in health information systems—especially given the gap left by reduced USAID investments. Nagpal also emphasized the need to reframe the conversation: “You should not be talking about digital health, but digital *in* health.” In other words, HIS must be fully embedded in system operations, not treated as a standalone innovation.

A related concern is the sustainability of any information system, as short-term donor projects can lead to limited national buy-in and poor long-term sustainability. Governments face financial and technical constraints in maintaining HIS initiatives once external funding ends, and many tools fall into disuse once donors exit, a challenge exacerbated by **poor internet infrastructure, unreliable electricity, and low digital literacy in rural areas** ([Serge et al., 2024](#), pp. 26, 19, 20). Indeed, pilot programs often don’t continue beyond their initial funding cycle ([Karamagi et al., 2022](#), p. 8). However, progress is being made: Nagpal pointed to some countries, like Indonesia and Lao PDR, which are making progress toward integration (using platforms like HL7 FHIR or DHIS2), and emphasized that integration and user-centered design must be core goals. “Use of an information system should not create more work... but actually reduce work,” he stressed. In DRC, Akilimali confirmed that **DHIS2 will remain in place despite USAID cuts**, but noted that components such as internet access, training, and system maintenance had been donor-funded and now face uncertainty.

Multiple experts, including Goldberg, Akilimali, and Ngongo, confirmed that USAID had funded the backbone of health data systems in much of SSA, and noted that USAID cuts will drastically impact HIS (see also [here](#)). As Ngongo put it, “Without [USAID] funding, many data collection systems will degrade,” particularly those that rely on periodic updates, internet access, or digital training programs. While some better-resourced countries may be able to absorb these costs, she cautioned that others (e.g. South Sudan, CAR, or parts of DRC) are at serious risk of system degradation.

As we discuss below, **neither an mHealth intervention nor an HR system tool appeared to meet cG’s bar**. However, given the emphasis placed on this topic by nearly all experts, it may be worth cG’s time to consider other interventions in this space that may have greater impact. We have also found some organizations providing technical support for national health information systems, including [Dimagi](#), [HealthEnabled](#), [Jembi](#), and [reach52](#), which could play a crucial role in maintaining and improving current infrastructure. We have not had a chance to look further into these organizations.

mHealth app and HR management system

In order to understand how information systems might perform compared to cG’s standards, we built two rough models on two different information interventions: one mHealth app to support community health workers (CHWs), and one HR management system.²⁹ We chose these as two different paths to impact for information technology: one directly improving the quality of care delivered, and one ideally increasing the effectiveness of the health workforce.

[D-Tree](#), the organization that produces the mHealth intervention we modeled, provides decision support and care integration for CHWs. Digital decision aids such as mHealth apps are often touted as a way to improve CHW efficiency, reduce diagnostic errors, and increase adherence to clinical guidelines. We chose this system to model because D-Tree claims that their app is highly interoperable with other systems, meaning that it might be a good candidate

²⁹ Both efforts are relevant in multiple “building blocks:” mHealth apps are tightly integrated with service delivery and task-shifting interventions, and HR management is closely tied to workforce interventions.

to improve CHW services without introducing further fragmentation. We think our model is most useful as a framework for estimating the potential costs and benefits of any generic open-source mHealth app, as we did not find a specific study attesting to the effectiveness of a single tool. Our very rough cost-effectiveness analysis suggests that while such tools may lead to incremental improvements in care quality, their impact on mortality and morbidity may be too small to justify prioritization. Our [rough model](#) showed an mHealth intervention significantly underperforming versus cG's bar at 503x.

An HR management system (in this case, we chose to model [iHRIS](#), a product developed by [IntraHealth](#) for USAID projects) showed somewhat greater promise, with a [rough cost-effectiveness estimate](#) (1,285x) somewhat closer to cG's funding bar. This suggests that improving HR or workforce management, particularly through better deployment, scheduling, and tracking of health workers, may be a more effective use of digital health tools than frontline clinical decision aids. The key question remains whether HR management systems can actually cause efficiency gains in terms of increased availability of doctors and nurses, and whether such gains can meaningfully translate into improved patient outcomes. While iHRIS appears to offer benefits in terms of reducing workforce bottlenecks, its impact likely depends on the degree to which better staffing allocation leads to higher service utilization and quality.

Financing

Our previous HSS report identified financing as an important but challenging area, with mixed evidence on existing interventions ([Kudymowa et al., 2025](#)). Country case studies highlighted the role of reducing out-of-pocket expenditures in improving outcomes, but most financing reforms (e.g., performance-based financing, health equity funds, insurance) appeared to have either uncertain cost-effectiveness or significant adverse effects.³⁰ Akilimali also pointed to **financial mismanagement and governance failures as critical barriers to effective health system improvements**. He advised for strong financial governance and earmarking measures to ensure funds are used as intended.

Nagpal, Goldberg, and Ngongo all emphasized that **financial semi-autonomy**³¹ and **community-focused incentive alignment have been essential** to the success of health reforms. Nagpal highlighted results-based financing successes in Laos, Cambodia, and Indonesia, where **small performance-linked grants led to rapid behavior change**. He emphasized that incentives were effective only when providers had some control over local budgets and predictable disbursements. These reforms aligned provider behavior with health priorities and improved care quality at modest cost.

Experts further argued that giving **providers the flexibility to allocate funds, hire staff, and respond to local needs** (within clearly defined rules) can generate substantial efficiency gains and service quality improvements. Nagpal noted that where providers lacked discretion over resources or where payment systems were opaque, financial incentives often failed to produce results. Goldberg similarly pointed to evidence that **greater worker and facility autonomy**, particularly in hiring, firing, and pay, can improve care quality and productivity. He noted that

³⁰ However, our further exploration of health information and digital interventions highlighted several potentially interesting microfinance organizations that may be worth considering further, including [10mg](#), an AI-powered credit scoring platform facilitating loans for healthcare providers in emerging markets, predominantly in Africa, and [reach52](#), which provides microinsurance, microfinancing, and access to affordable medicines in rural and peri-urban areas in Asia and Africa.

³¹ This type of financial semi-autonomy may conflict with recommendations like Akilimali's to earmark funds for specific uses. We did not explore this tension in depth but note it remains an open debate.

when public primary care facilities had the same level of autonomy as private ones, quality gaps disappeared. Ngongo shared an example from Senegal, where RTI's USAID-funded health program linked grant disbursements to government performance milestones validated by regional committees. The model emphasized government ownership and community accountability, supporting her view that donors should empower national systems to set and pursue their own priorities.

Results-based financing

We also created a [rough cost-effectiveness model](#) on Results-Based Financing (RBF) for health centers, modeled on a component of a [USAID \(2015\)](#) Integrated Health Program. In that program, financial incentives in seven "health zone" districts in the DRC were tied to predefined performance indicators, mainly related to maternal and child health, service utilization, quality of care, and management. Community scorecards (CSCs) and broader community accountability mechanisms were embedded in the program, primarily through the CODESA structures (Comités de Développement Sanitaire), which played a central role in community verification and social accountability. The program has been credited with considerable improvements in service utilization, facility quality, and community engagement, with assisted deliveries increasing from 18% to 90% and health center functionality scores rising significantly.

Our model suggests that while **financial incentives can improve behavior and service delivery, their cost-effectiveness (1,305x) likely falls short of cG's bar**. Although we previously saw RBF as a promising way to improve spending efficiency, our updated analysis indicates the impact may be too limited to warrant prioritization. This also has **implications for community scorecards (3,123x)**, which we previously highlighted and are also sometimes embedded in RBF programs ([Kudymowa et al., 2025](#), p. 30). If RBF underperforms, it raises questions about whether CSCs meet cG's bar or if they are more effective as standalone governance tools rather than financial accountability mechanisms.

Leadership and governance

Our prior report suggested governance reforms could be highly promising, particularly in fostering long-term systemic change ([Kudymowa et al., 2025](#)). Supervision and mentorship for IMCI providers appeared extremely promising, potentially surpassing 8,000x. We also analyzed community scorecards (CSCs) as an accountability mechanism and found that their cost-effectiveness (~3,000x) may exceed cG's bar (see the prior section on [Results-Based Financing](#) for an update on this analysis). However, we also found in our prior report that continuous quality improvement (CQI) may be less cost-effective, at less than 500x ([Kudymowa et al., 2025](#), p. 37).

Experts in our prior phase of research suggested instead focusing on public financial management reforms, particularly improving procurement, transparency, and budget predictability. Advocacy efforts to align these financial reforms, whether centralizing or decentralizing, whether privatizing or bringing into the public sector, came up repeatedly in our prior work as an area for further exploration. We have since found the [African Health Budget Network \(AHBN\)](#), [Health Economics and Epidemiology Research Office \(HE²RO\)](#), and [Management Sciences for Health \(MSH\)](#) as organizations that claim to do advocacy work along these lines, although we have not looked into this work further.

Expert interviews during our prior research also supported the importance of long-term investment in think tanks, universities, and advocacy groups that can push for evidence-based

health reforms, which we also discuss below for “[research](#)” interventions. We have identified several organizations pushing for policy reform, including those focused on health financing and regulatory improvements,³² but our initial searches did not find an obvious candidate for a rough cost-effectiveness model.

While investing in government institutions is valuable, our further research indicates that **large-scale, broad, or less-targeted governance reforms tend to be slow-moving and difficult to evaluate**, as discussed [above](#). Nagpal stressed that governance reform is most tractable when there is political momentum and committed champions. He cited Indonesia’s current health reform drive as a rare window of opportunity. This suggests that governance-focused interventions may be most impactful when timed to coincide with political reform cycles, rather than delivered as a generic technical package. Our own research indicates that donor-funded governance initiatives are sometimes short-term and project-based (see the [Gavi case study](#)), further highlighting the need for local stakeholder support. Nagpal also highlighted the importance of financial governance and decision-making autonomy.

Goldberg offered a nuanced view on CSCs and local monitoring tools, which we had previously flagged as promising based on early-stage evaluations. He acknowledged that some advocates view community monitoring as a powerful way to “break the vicious cycle” of patient mistrust and provider absenteeism, but cautioned that there may be strong arguments in favor of skepticism. He noted that while community-based programs often show strong pilot results, there are signs that they struggle with long-term sustainability and scale.

Health workforce

Our past research on the health workforce focused on CHWs as a representative intervention for task-sharing and service extension ([Kudymowa et al., 2025](#), p. 27). While CHWs improved health coverage, they did not appear to be sufficiently cost-effective for cG. Our expert discussions emphasized workforce conditions over expansion, with suggestions to improve job satisfaction for human resources professionals and invest in local research talent rather than relying on international experts.

In this phase of the project, we have largely set aside health workforce interventions, except to document organizations that warrant future consideration, such as [D-Tree International](#) and Medic Mobile ([Devex, 2009](#)). However, our [rough model](#) of a hypothetical HR-management software tool, while not meeting cG’s bar, was one of the most promising of the interventions we reviewed for this stage of the report at 1,285 x.

Our case studies, of vertical funders such as [Gavi](#), and vertical programs such as the [Malaria Consortium’s SMC work](#), have highlighted that the health workforce is an extremely important but difficult area of health systems. Indeed, vertical programs are often criticized for exacerbating workforce limitations through competition, and low pay and limited career advancement opportunities lead to high attrition rates in the field.

³² These include [Resolve to Save Lives](#), which works on policy advocacy for non-communicable diseases and primary health system reforms; the [Global Initiative for Economic, Social & Cultural Rights \(GI-ESCR\)](#), which advocates for equitable public health financing and governance transparency; and the [Public Health Resource Network \(PHRN\)-India](#), which focuses on health policy reform and governance advocacy.

Service delivery

In our prior report, we explored service delivery interventions such as Integrated Management of Childhood Illness (IMCI) ([Kudymowa et al., 2025](#), p. 22). We found that IMCI, while widely implemented, could still benefit from supervision and mentorship, which appeared to have a cost-effectiveness exceeding cG's 2,000x bar (*ibid*, p. 37). We also examined broader service integration efforts but found that experts did not emphasize them as a key route for health system reform.

We deprioritized this building block in our current research phase given its strong coverage in the previous round, but expert interviews reinforced several key ideas that may be worth revisiting. In our conversation, Nagpal argued that the root problem in many underperforming facilities is often not workforce quantity, but productivity, readiness, and access to necessary equipment. He noted, however, that in some cases— particularly for more specialized or better-trained cadres— staff shortages also contribute meaningfully to poor performance.

Research

We have not explored standalone health systems research funding in depth, particularly since “research” is both a very broad term and is not included as a WHO building block. However, expert interviews during the previous project suggested that investing in local research talent, particularly in epidemiology and implementation science, could strengthen health systems. These initial conversations also raised that longitudinal tracking of system-strengthening grants is often neglected, making it difficult to assess their true long-term impact. During the most recent stage of our HSS project, experts across institutions emphasized that **health systems research is unusually neglected, increasingly urgent, and high-impact in the wake of the USAID withdrawal**. They identified multiple places where targeted funding could generate actionable insights for policymakers and implementers alike, and close time-sensitive gaps.

Several experts recommended various types of meta-research to further identify which health systems interventions work, and how they interact with local systems. For example, Ngongo noted that non-systems interventions like **Anders Seim's postpartum hemorrhage initiative in Niger** showed incredibly impressive results despite the authors not focusing on strengthening systems. She suggested research could explore how such programs can complement or evolve into systems-oriented efforts. In addition, both Nagpal and Goldberg highlighted the [Joint Learning Network for UHC](#) as a **high-leverage peer-learning platform, reaching 40+ countries at low cost**. This network could double as a channel for piloting research, collecting comparative data, or testing implementation strategies across contexts.

Adam Salisbury is reviewing GiveWell's TA grants, about 10% of their recent portfolio. He highlighted challenges in assessing counterfactuals and success rates, and noted that long-term impact evaluations for HSS remain underfunded. He also expressed interest in better tools for measuring impact and in researching local policymaker views on vertical vs. systems programs, potentially with Mattie Toma's [Policymakers Lab](#).

Ngongo, Akilimali, and Salisbury all stressed the fragility of and importance of maintaining high-quality data ecosystems in LMICs, both to inform research and to help with strategic decision-making. One health expert in Francophone West Africa similarly emphasized that “the first priority is strengthening health information,” citing severe gaps in epidemic surveillance, cause-of-death data, and real-time data for decision-making. They described examples where misallocated health personnel and investments resulted directly from missing

data: “We invested heavily in a region without having data on how many doctors or nurses were there. Other regions were actually more in need.”

Several experts raised concerns about the loss of **USAID support for the Demographic and Health Surveys, a cornerstone of global health evidence** that underpins strategy and research work in dozens of countries. Ngongo called its interruption a “huge loss,” noting that without DHS it will be challenging to determine the impacts of USAID funding cuts. As of August 2025 it was unclear which components of DHS might be able to continue, either through country ownership/funding or paid by other donors. Akilimali says he participates in [Countdown 2030](#) using both DHIS2 and DHS data to support policy feedback loops, and noted that DHS data “wouldn’t necessarily be standardized worldwide” if countries attempted to replace it piecemeal. It’s possible that targeted support could at least extend data collection on DHS surveys in particular countries to minimize data disruption, although the scale and cost³³ of DHS surveys more broadly would be impossible for OP-scale grantmaking to replace.

We have further identified a large number of organizations that are involved in HSS research of some kind, although we have not had a chance to validate any of these organizations with experts and have not built models of their potential effectiveness. This includes many that do research to inform LMIC health policy, such as the [African Population and Health Research Center \(APHRC\)](#), [Council on Health Research for Development \(COHRED\)](#), [Health Economics and Epidemiology Research Office \(HE²RO\)](#), [Health Systems Trust \(HST\)](#), [Innovations in Healthcare](#), [RESYST](#), and [Access to Medicine Foundation](#). We also identified research organizations that look further into service delivery and direct healthcare models, such as [ReBUILD for Resilience](#), [icddr.b](#), and the [George Institute for Public Health](#), as well as [Health Economics and Epidemiology Research Office \(HE²RO\)](#) and the [Health Systems Trust \(HST\)](#).

Rough geographic prioritization based on the INT framework

In this section, we conducted a rough prioritization of SSA countries where cG might focus its efforts on HSS using the importance, neglectedness, tractability (INT) framework.

To assess importance, we ranked countries based on estimated deaths amenable to healthcare ([Figure 1](#)), which provides a measure of the potential impact of strengthening health systems. To assess neglectedness, we ranked countries based on HSS funding received as a ratio of total DALY burden ([Figure 4](#)), indicating where HSS efforts are relatively underfunded. We then identified the top 20 countries for each metric and selected the overlapping countries between the two rankings. The top countries according to importance and neglectedness are: Cameroon, Central African Republic, Chad, Guinea-Bissau, Mali, Somalia, Zambia, and Zimbabwe. To incorporate tractability into the prioritization, we needed an indicator that reflected the feasibility of implementing and scaling HSS efforts in each country. We were unable to identify any single indicator that fully captures tractability, as it depends on multiple factors, including governance, health system capacity, external funding, and fragility.

Due to time constraints, we did not conduct a full tractability assessment in this report. However, we recommend that cG could further explore this dimension by:

- Consulting HSS experts with a good understanding of local contexts.
- Ranking countries using indicators such as:
 - Fragile States Index (FSI; [Wikipedia, 2025a](#)) for political stability. By skimming the 2023 Fragile States Index map ([Wikipedia, 2023b](#)), we can immediately see

³³ A single DHS survey in one country costs ~\$1.3M ([Kilic et al., 2017](#)).

- that the Central African Republic, Chad, and Somalia are either ‘failed’ or ‘high alert’ states, so these may have very low tractability.
- Government Effectiveness Index for policy implementation capacity ([Wikipedia, 2024a](#)).
- Health Service Coverage Index for system functionality ([WHO, 2024b](#)).
- Presence of major HSS donors (e.g., Global Fund RSSH, Gavi HSS, World Bank projects) as a proxy for external support.³⁴
- Reviewing past HSS programs to assess feasibility in different contexts.

At a relatively late stage of this report, we came across two other relevant pieces of information that could potentially inform geographical prioritization. We did not have sufficient time to investigate those further:

- Ngongo highlighted that French-speaking West Africa has a particular mismatch between population health needs and external donor support, though we note that this isn’t clearly reflected in [Figure 4](#).
- A recent analysis from the Center for Global Development (CGD) reveals the top 26 countries that are most exposed to US global health aid cuts. It also contains information on the fragile/conflict-affected status of those countries, which can be useful for the tractability assessment ([Baker et al., 2025](#)).

Case studies in HSS: Lessons from funders and implementers

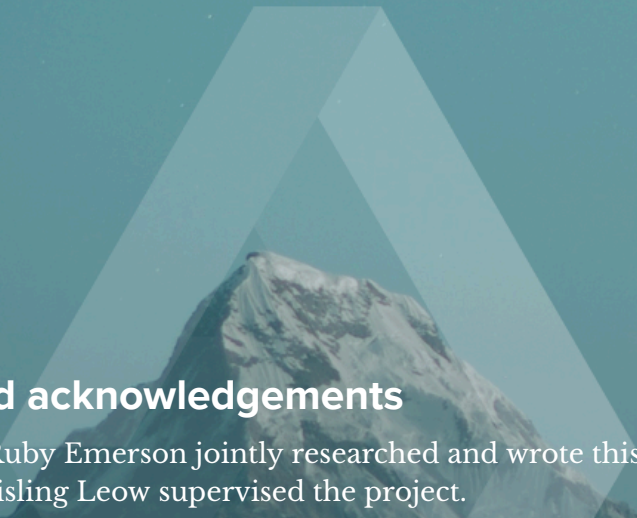
To better understand the opportunities and challenges of HSS, we reviewed six case studies: two focused on how large funders shifted toward HSS, and four examining how implementers integrated vertical programs into broader systems. See [Appendix J](#) for the full case studies.

On the [funder side](#), both [PEPFAR](#) and [Gavi](#) moved toward HSS largely out of concern for long-term sustainability. Their efforts, however, were shaped by political and financial constraints and faced implementation challenges. PEPFAR’s integration into national health systems showed greater success in high-capacity settings, while Gavi’s HSS investments often remained narrowly focused on immunization-related inputs. Both cases highlight the difficulty of embedding systemic change into vertical program structures.

Among [implementers](#), integration efforts showed mixed results. The [African Programme for Onchocerciasis Control \(APOC\)](#)’s community-based drug distribution model successfully expanded reach and supported other health efforts, but depended on unpaid labor and external funding. [Malawi’s integration of HIV, tuberculosis, and maternal and child health services](#) improved uptake but faced persistent challenges such as poor follow-up and workforce constraints. [Africa AHEAD’s Community Health Clubs](#) helped promote healthy behaviors and public health engagement at the community level, though long-term institutional impact remains uncertain. Finally, the [Malaria Consortium’s Seasonal Malaria Chemoprevention](#) campaigns, while aligned with government health worker structures, remained reliant on donor-run logistics and were vulnerable to political disruption.

Overall, these cases suggest that HSS and integration efforts can be impactful, but only under the right conditions. Sustained government buy-in, adequate implementation capacity, and long-term funding are often necessary for success.

³⁴ Higher existing donor engagement could mean that a stronger health system infrastructure is already in place, making implementation more feasible. It may also indicate opportunities for alignment with ongoing initiatives, co-funding, or leveraging existing institutional capacity. However, in some cases, high donor presence can lead to inefficiencies or coordination challenges, which should also be considered.



Contributions and acknowledgements

Jenny Kudymowa and Ruby Emerson jointly researched and wrote this report. Kudymowa also served as project lead. Aisling Leow supervised the project.

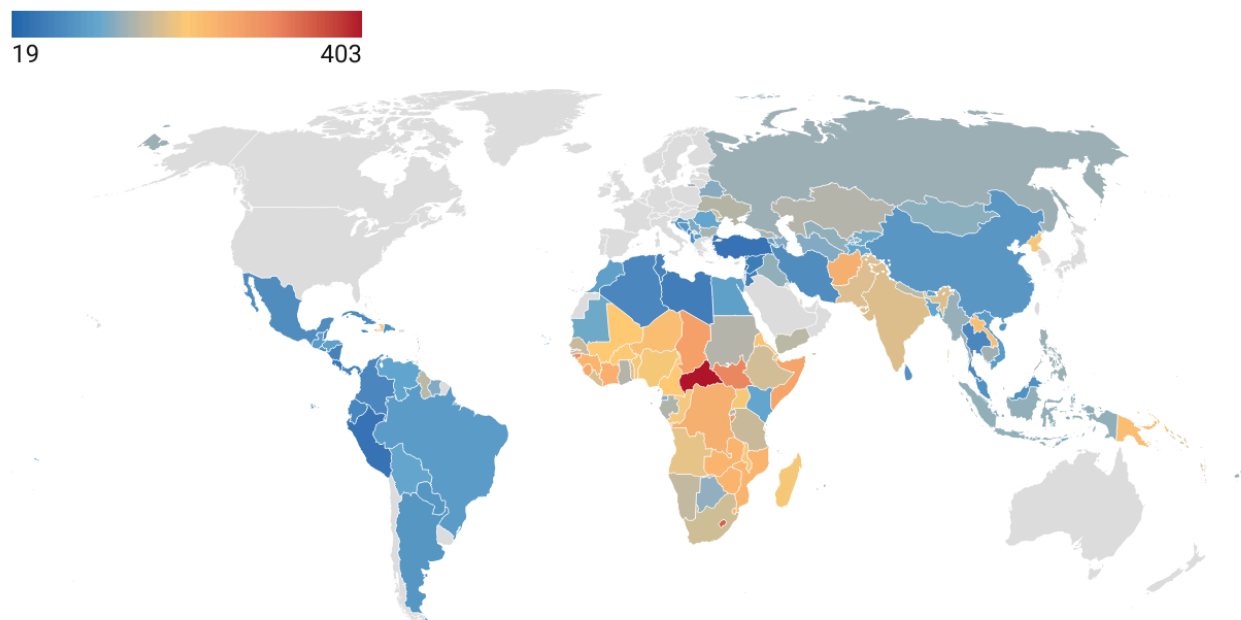
Special thanks to Aisling Leow, Greer Gosnell, Dylan Collins, Rafael Latham-Proença, and Rossa O'Keeffe-O'Donovan for helpful comments on drafts. Thanks also to Thais Jacomassi and Shane Coburn for copyediting and Sarina Wong for assistance with publishing the report online. Further thanks to Carrie Ngongo, Adam Salisbury, Pierre Akilimali, Jonah Goldberg, Somil Nagpal, and other anonymous experts for taking the time to speak with us.

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Appendices

Appendix A. Supplementary figures

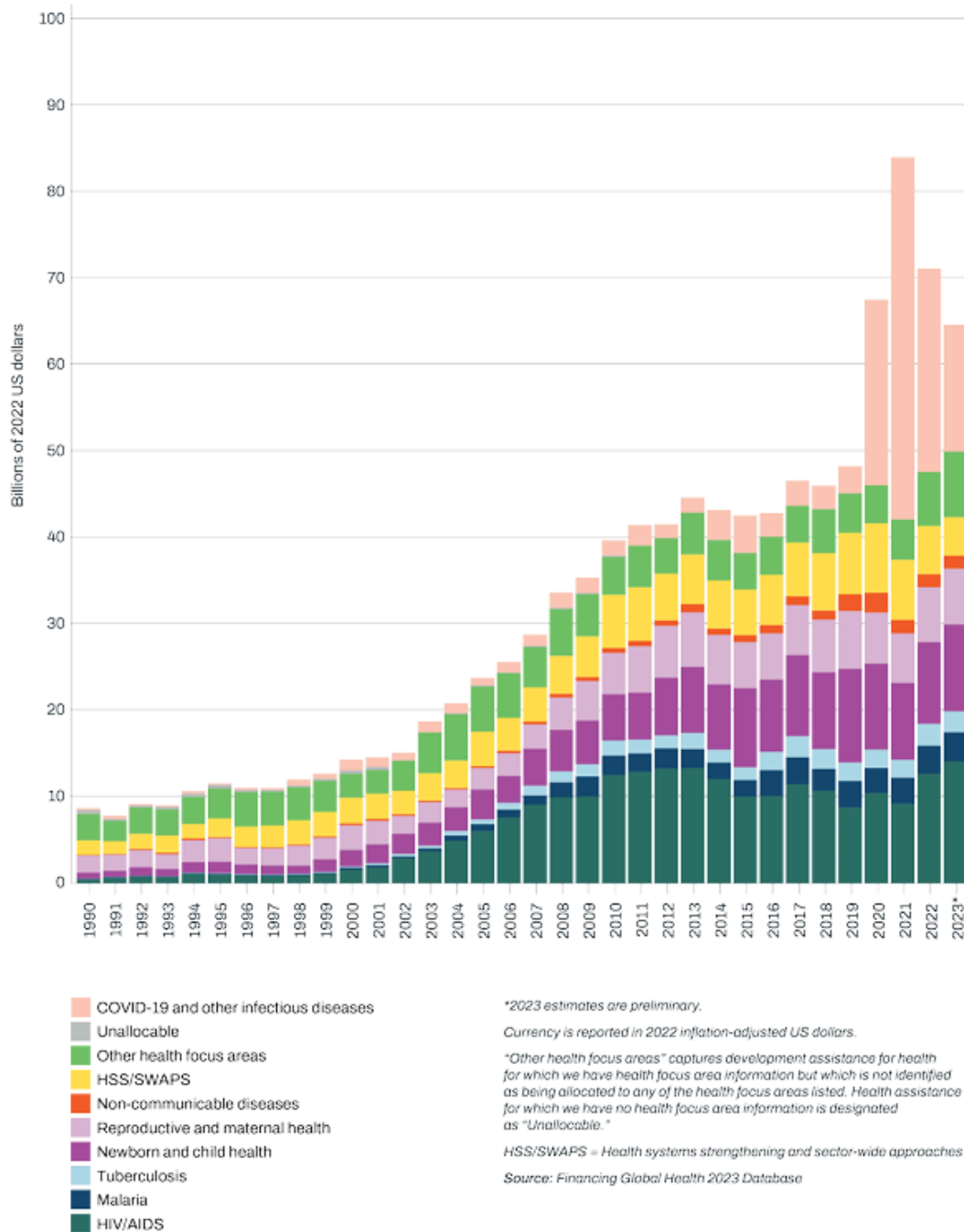
Figure A1: Deaths amenable to healthcare per 100,000 people across 137 low-and middle-income countries in 2018



Created with Datawrapper

Note. Calculated based on Table S5 in [Kruk et al.'s \(2018a; 2018b\)](#). Calculations [here](#). Interactive visualization on [Datawrapper](#).

Figure A2: Development assistance for health by health focus area, 1990-2023



Note. Figure 10 from [IHME \(2024a\)](#).

Appendix B. Methodology in Kruk et al. (2018)

Below is a summary of the methodology used by [Kruk et al. \(2018a\)](#) to estimate excess deaths due to low-quality health systems (amenable mortality) in 137 LMICs:

Estimation of excess mortality for 62 diseases in LMICs:

- The authors first excluded deaths that could have been prevented by population-level interventions (such as public health measures) to focus only on deaths that healthcare systems could prevent. Concretely, they excluded deaths that could have been prevented by public health policies, vaccination programs, sanitation improvements, or lifestyle changes. For example, deaths from lung cancer due to tobacco control failures were considered preventable at the population level, not amenable to personal healthcare. Included deaths were those requiring direct medical care, e.g., treating a heart attack or managing childbirth complications.
- After excluding those deaths, the authors considered 62 health conditions that are amenable to healthcare (i.e., conditions that are known to be treatable with medical interventions) and relied on data from the 2016 Global Burden of Disease (GBD).
- They estimated excess mortality by comparing case fatality rates (CFRs) in 137 LMICs with those in 23 high-income reference countries that have strong health systems. The difference in CFRs between LMICs and reference countries was used to calculate the number of deaths that could have been prevented with high-quality healthcare.

Categorizing excess mortality into non-utilization and poor quality of healthcare:

- The authors categorized excess mortality into two groups:
 - **Non-utilization deaths:** deaths among people who never sought medical care or did not have access to healthcare.
 - **Poor-quality deaths:** deaths among people who did seek care but still died due to low-quality treatment or ineffective healthcare services.
- They assumed that if those who accessed healthcare (based on household survey data from e.g., the Demographic and Health Survey) had received high-quality services, their CFRs would match those observed in high-income reference countries. Any excess deaths among care-seekers were attributed to poor healthcare quality.

Appendix C. Methodology and limitations in IHME's estimation of HSS funding

IHME's estimates for HSS funding streams are made in two stages (see Financing Global Health 2023 Methods Annex [[IHME, 2024b](#)] in Sections 1.3 and 3):

1. **Data collection from various sources:** DAH data are drawn from a variety of sources, including financial statements, annual reports, budget documents, and project disbursement records from agencies like the OECD Creditor Reporting System, World Bank, Global Fund, and major philanthropic foundations. Some sources are obtained through personal correspondence, and various checks are made to avoid, e.g., double-counting of funding streams (p. 8).
2. **Disaggregation into health focus areas using keyword searches:** DAH spending is then disaggregated into health focus areas using keyword searches of project titles and descriptions. Total DAH is then split across focus areas based on weighted averages of keywords; for instance, if three keywords indicate HIV/AIDS and two indicate HSS/SWApS, three-fifths of the DAH is allocated to HIV/AIDS and two-fifths to HSS/SWApS (p. 39).

Based on a high-level review of the methodology, we see two main limitations:

1. The data set excludes funding from foundations outside of the US.³⁵ Thus, total DAH funding is likely underestimated, though we are unsure by how much. We think that the Gates Foundation is likely the largest individual foundation involved in HSS, so we do not expect that other foundations outside of the US play a very major role, but we have not reviewed foundations outside of the US besides the Wellcome Trust.
2. The terms used in the keyword search ([IHME, 2024b](#), Table S3.4) seem to be predominantly centered around human resources and do not encompass terms from all six HSS building blocks as defined by the WHO. For example, the term “supply chains” does not seem to have been used. Thus, we suspect that DAH funding for HSS/SWApS may be underestimated relative to other health focus areas.

³⁵ “Previous studies on foundations outside the US have documented the severe paucity of reliable time series data and lack of comparability across countries. Hence, this research focused efforts on tracking only US foundations.” ([IHME, 2024b](#), p. 78).

Appendix D. Methodology in Stenberg et al. (2017)

[Stenberg et al. \(2017a\)](#) project health system resource needs for 67 LMICs from 2016 to 2030. The authors categorize health services into four delivery platforms: community-based health services, health centers, first-level hospitals, and referral hospitals. Costs and health outcomes are estimated based on projected service scale-up, workforce needs, and infrastructure investment.

They model two scenarios to evaluate different levels of ambition in achieving health targets:

1. **Progress scenario:** Assumes that countries' advancements toward global targets are moderated by their health systems' absorptive capacities, leading to a more gradual improvement.
2. **Ambitious scenario:** Posits that most countries will attain the global health targets by 2030, requiring a more rapid and substantial scale-up of health services.

For each scenario, the study estimated the associated financial costs and projected health outcomes, including assessing reductions in disease prevalence, lives saved, and increases in life expectancy resulting from the proposed health interventions. The authors projected available funding for health by analyzing economic growth trends and anticipated health sector allocations in each country. The methodology relies on modeling projected service delivery expansion, workforce and infrastructure needs, and intervention costs, while incorporating country-specific economic and demographic factors to estimate feasibility and impact.

The methodology has a number of **key limitations**, e.g.:

- **Uncertainty in projections:** The estimates for health system resource needs rely on assumptions about future GDP growth, government health allocations, and external funding availability, all of which are highly uncertain, particularly in fragile and conflict-affected states.
- **Uncertain health system absorptive capacity:** The model assumes that health systems can efficiently absorb large increases in funding and infrastructure investments, but weak governance, corruption, and workforce shortages may hinder implementation.
- **Limited scope of included health interventions:** The study only models interventions with well-established effectiveness and available data, meaning that some potentially important but under-researched interventions are excluded. Examples of excluded interventions are suicide prevention programs, certain cancer treatments, and emerging therapies for non-communicable diseases (NCDs). This could lead to an underestimation of the resources required for comprehensive health system strengthening.
- **Exclusion of certain health needs:** The study does not include some key health conditions, such as road traffic injuries, hepatitis treatment, and chemical poisoning, due to gaps in available data.
- **Limited country-specific customization:** The projections apply broad, generalized assumptions about service scale-up and cost structures across 67 LMICs, which may not fully reflect country-specific health system realities.

Appendix E. Existing vs. required allocation of HSS funds across building blocks

Table E1: Existing vs. required allocation of HSS funding across building blocks

Health system building block	Existing allocation of HSS funding in 2015 according to Kraus et al. (2020, Figure 4)	Required allocation of HSS funding by 2030 according to Stenberg et al. (2017a)
Service delivery	38.5%	35.9%
Governance	20.3%	0.7%
Health workforce	13.5%	56.2%
Health financing	13.3%	1%
Health information systems	7.7%	0.2%
Supply chain	6.7%	6.2%
Emergency risk management	Not estimated	0.8%

Note. Required allocation calculated based on estimates in Table S16 in [Stenberg et al. \(2017a; 2017b, p. 56\)](#). We only considered the allocation of funding across HSS building blocks here, not the allocation across horizontal vs. vertical health programs.

Appendix F. Overview of major implementing organizations involved in HSS

[Table F1](#) below shows an overview of eight of the major implementing organizations involved in HSS. We identified these organizations based on a combination of prior knowledge and ChatGPT queries. We are highly uncertain about whether these are in fact the largest HSS implementers, and our annual HSS spending estimates should be interpreted with caution, as they are based on rough guesstimates and very limited publicly available data. None of the organizations we reviewed provide a clear, disaggregated accounting of their HSS-specific expenditures. We therefore estimated their HSS spending by starting from total annual expenditures and applying rough assumptions about the proportion of their work focused on HSS, based on skimming their websites and annual reports.

Table F1: Overview of major implementing organizations involved in HSS

Organization	Annual HSS spending	Key HSS focus areas and activities	Assumed share of annual program spending on HSS
Partners in Health (PIH)	~\$150M ³⁶	• Five S model: staff, stuff, space, systems, and social support (PIH, 2021). • Focus areas include community health workers, health workforce training, social support services, and surgical care. • They also engage in research, advocacy, and policy influence.	~70%
Management Sciences for Health (MSH)	~\$100M ³⁷	• They seem to be involved in all HSS building blocks and emphasize universal health coverage. • Activities: e.g., technical assistance to health ministries, policy and reform, capacity building, data systems design and use, management and governance training, health financing strategy development	~75%
Abt Global	~\$100M ³⁸	According to Abt Global (2024) : • Health policy development and evaluation • Financing, resource mobilization, and optimization • Human resources for health • Leadership, management, and governance • Domestic resource mobilization • Private sector engagement	~25%

³⁶ In 2024, PIH spent \$245M in total, 90% of which was program spending ([PIH, 2024](#)). As HSS seems to be a core focus of their approach, we guess that ~70% of their spending goes towards HSS, i.e., ~\$150M.

³⁷ ~\$130M total program expenses in 2023 ([MSH, 2024](#)). We did not find any specific figures on HSS expenses, but as MSH seems to be heavily specialized in HSS, our guess is that ~70-80% of its portfolio goes towards HSS, i.e., ~\$100M.

³⁸ We have not found their financials in a quick search. According to Management Consulted (2024), ([Management Consulted, 2024](#)), Abt Global's total annual revenue is ~\$400M. Our guess based on skimming the [Abt Global](#) website is that a quarter of their funding goes to HSS, i.e., ~\$100M.

		Digital and information systems solutions	
Clinton Health Access Initiative (CHAI)	~\$90M ³⁹	- Mainly focused on health financing and health workforce (CHAI, 2023). - Health financing: supports governments with benefit package design, budgeting, and public financial management through technical assistance and policy modeling. - Health workforce: support with workforce planning, training, and HR data systems via embedded advisors and capacity-building support.	~40%
Jhpiego	~\$90M ⁴⁰	- Under the umbrella of HSS, they work on primary health care, global health security, nursing & midwifery, and safe surgery. - Within primary health care (Jhpiego, 2021), they seem to be active across all HSS building blocks, but we did not find detail on specific activities.	~25%
FHI 360 (formerly Family Health International)	~\$70M ⁴¹	- The website (FHI 360, 2013) is very vague about their HSS involvement, emphasizing local leadership, resilience, and systems thinking—while offering limited detail on specific activities. - A recurrent theme is that they integrate disease-specific programs into broader service delivery systems.	~10%
John Snow, Inc. (JSI)	~\$70M ⁴²	- The JSI (2019) website mentions HSS as a core area of their international work without any specific detail on focus areas or activities. - They seem to be involved in health information systems.	~20%

³⁹ CHAI spends \$226M per year in total. ([CHAI, 2023](#)). They don't indicate how much they spend on HSS. While they originally started as a more vertical-focused organization, our impression is that they considerably scaled their HSS involvement in recent years. Thus, our guess is that ~40% of their funding goes towards HSS, i.e., ~\$90M.

⁴⁰ In 2022, Jhpiego received \$360M in funding ([Jhpiego, 2023](#)). We have not found any indication of their expenses in a 5-minute search. Our impression is that the core of their work is on vertical programs, but some of their work is on HSS. Our guess is a quarter, so ~\$90M.

⁴¹ [FHI 360's \(2024\)](#) financials state that operational expenses (which we think are program expenses) were ~\$714M. We think that roughly half of their spending is on health, and guess that ~20% of that is for HSS, i.e., \$70M.

⁴² We have not found a JSI financial report, but, according to [ProPublica](#), 2023 expenses were \$322M. Our guess is that they spend ~70% on their LMICs work and the remainder in the US. HSS seems to mainly support vertical programs, but our guess is that 30% of their work in LMICs is on HSS, i.e., ~\$70M.

PATH	~\$60M ⁴³	• Focus on digital health, data systems, and infrastructure to improve service delivery. • Provides technical assistance on policy, financing, and workforce planning.	~20%
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⁴³ ~\$318M total program expenses in 2023 ([PATH, 2023](#)). HSS is not a standalone budget line and we found no indication of how much they spent on HSS. Our guess based on skimming the website is that ~20% of their expenses go towards HSS, i.e., ~\$60M.

Appendix G. Overview of health product supply chains in LMICs

How are health product supply chains organized in LMICs?

The best overviews of how health product supply chains are organized in LMICs we found are [Yadav \(2015\)](#) and [Pisa and McCurdy \(2019\)](#). We summarize these in this section.

Unlike in high-income countries, where health supply chains are mainly managed by private companies, LMICs generally rely on a mix of public, private, and NGO-led systems that interact in complex, multi-tiered systems (see [Figure G1](#) below for an overview of the structure of health supply chains in LMICs). Governments, often in collaboration with multilateral organizations like Gavi and PEPFAR, play a dominant role, but private sector involvement is increasing as income rises. The key components are:

1. Manufacturing & procurement

Medicines are produced by local or international manufacturers. Procurement is handled by governments, donors (e.g., Global Fund, PEPFAR, Gavi), NGOs, and private buyers. Some countries use centralized national procurement systems, while others have fragmented procurement at regional or facility levels.

2. Storage & national distribution

The most common model in LMICs involves a central medical store (CMS), which is a government-run agency that stores and distributes medicines nationwide. The CMS also manages a tiered network of regional and district medical stores. Some governments and donors maintain dedicated transport fleets, but some rely on third-party logistics providers.

3. Parallel donor supply chains

Many global health programs set up independent supply chains (e.g., for HIV, malaria, TB) rather than relying on weak national distribution systems. These vertical supply chains sometimes overlap with national systems (e.g., shared warehouses) but often operate separately to ensure rapid access to medicines. The integration of global health initiatives into national supply chains varies considerably across organizations.⁴⁴

4. Last-mile distribution & retail

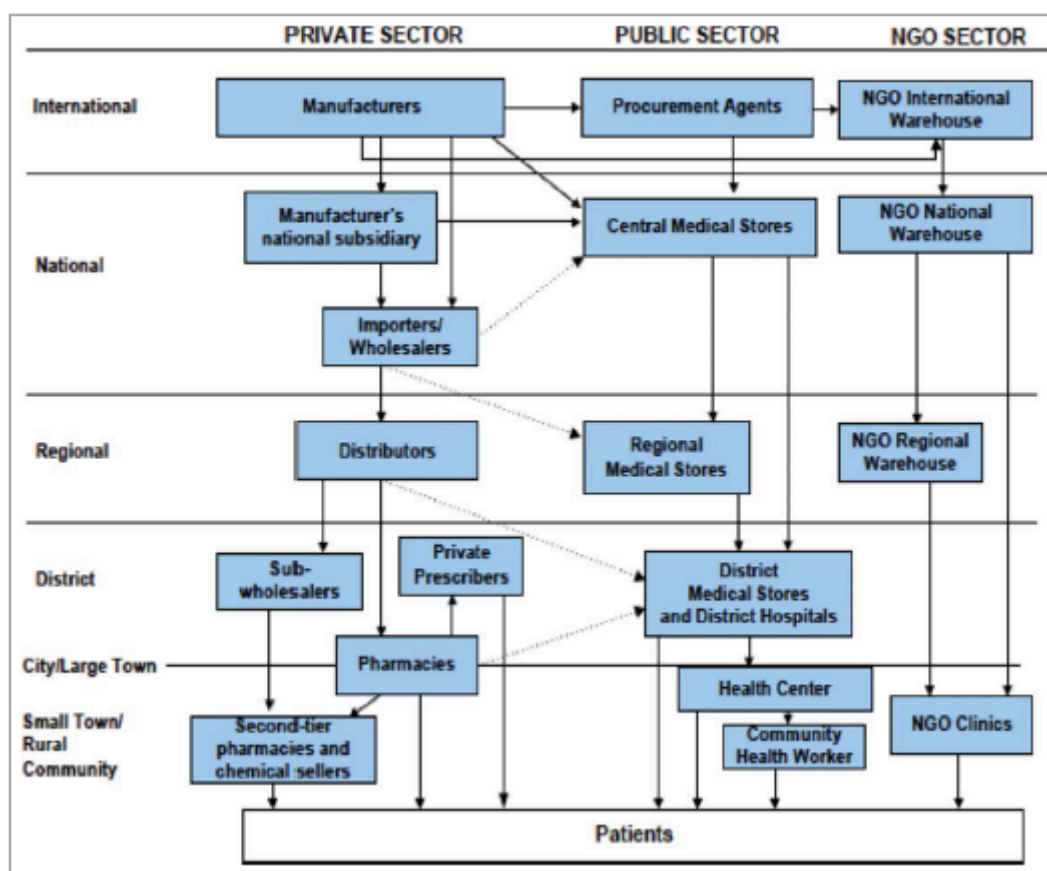
Medicines are delivered to health facilities, pharmacies, and clinics, typically through public sector supply chains or donor-supported programs. Private pharmacies and informal drug sellers play a major role, especially where public systems are weak. Some countries use community-based distribution models, where medicines are provided by trained health workers in remote areas.

These multi-layered systems can be incredibly complex⁴⁵ and opaque, with different levels of integration and coordination across countries. This makes them vulnerable to inefficiencies and a variety of challenges (e.g., corruption, product diversion, theft, falsification of drugs).

⁴⁴ “For example, PEPFAR has worked closely with local staff to strengthen national supply chains since its inception. In contrast, the Global Fund took a largely hands-off approach through its first 10 years of existence until it changed course in 2013. This shift followed the publication of an internal report that identified procurement, storage, and distribution of medicines as “significant vulnerabilities” posing major risks to the organization’s finances, operations, and reputation ([Pisa & McCurdy, 2019](#)).

⁴⁵ See Figure 2 in [Pisa and McCurdy \(2019\)](#) which maps the health supply system in the Democratic Republic of Congo, for an illustration of this complexity.

Figure G1: Structure of the health supply chain in LMICs



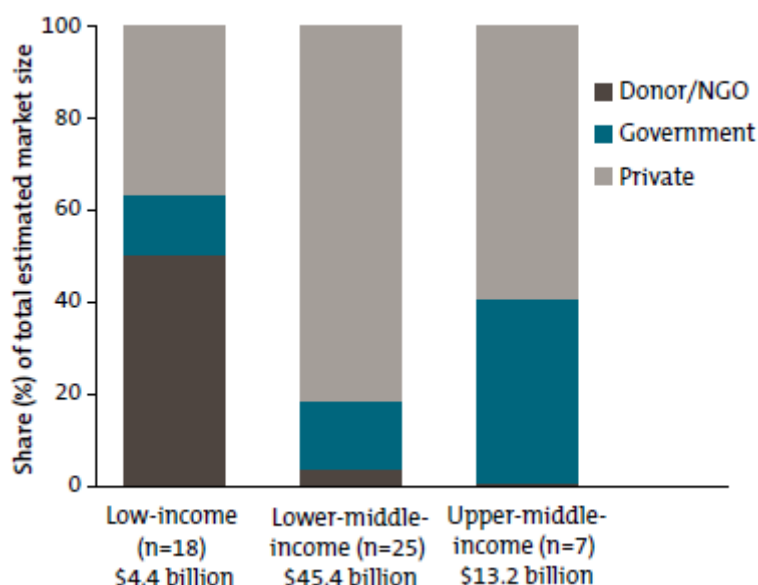
Note. Copied from [Yadav \(2015\)](#)

Who pays for drugs in LMICs?

In LMICs, the financing of drugs involves a combination of sources ([Silverman et al., 2019](#); see also [Figure G2](#) below):

- **Donor funding:** International donors, NGOs, and global health initiatives (e.g., the Global Fund, Gavi, PEPFAR) provide substantial financial support for medicine procurement and distribution, but the share of donor funding varies widely across country income levels. As income levels increase, governments and private sources take on a larger role. In low-income countries, donors account for ~50% of all health product expenditures. In lower-middle-income countries, donors account for only ~5%.
- **Private sector and out-of-pocket payments:** In lower-middle-income countries, about 80% of all health products are procured through the private sector, and individuals often pay for drugs out of pocket.
- **Government funding:** Governments in low-income and lower-middle income countries currently represent a small share of total purchasing for medicines and other health products, ~10-15%. They represent a much higher share, ~40% in upper-middle income countries.

Figure G2: Private, government, and donor/NGO financing as a share of the total estimated market value for health products by country income groups

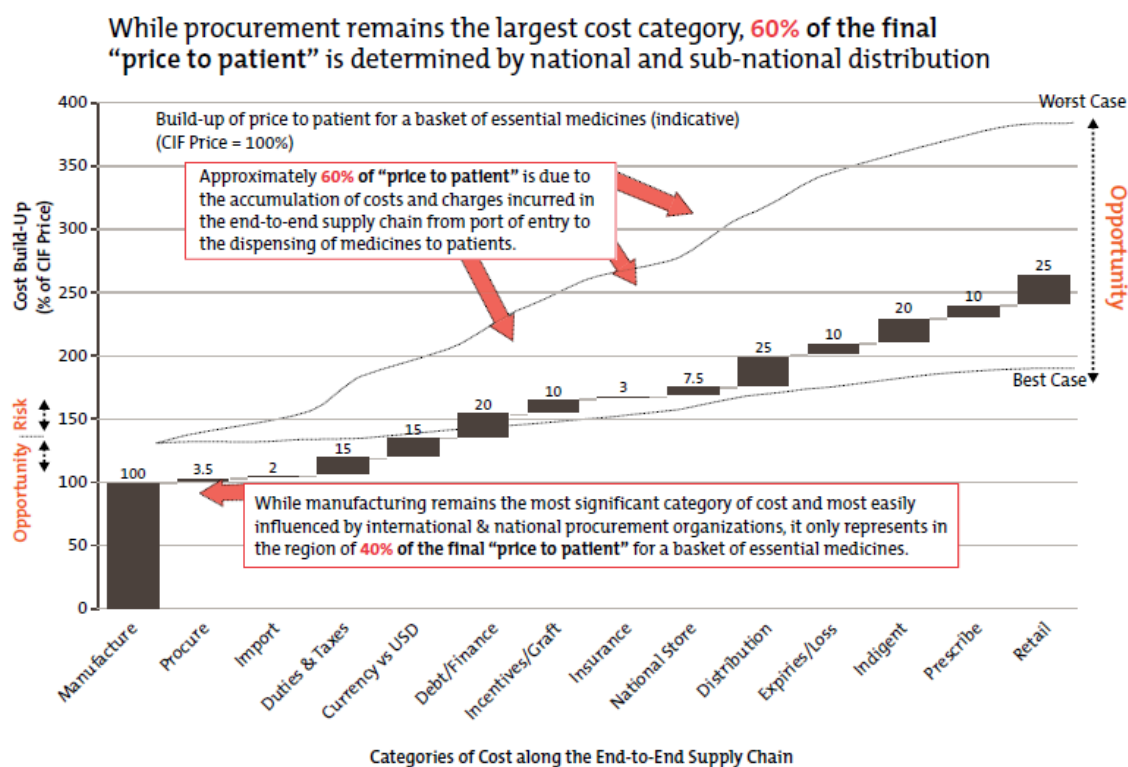


Note. Copied from [Silverman et al. \(2019\)](#)

What factors drive the costs of health products?

[Figure G3](#) below illustrates how mark-ups along the health supply chain significantly impact final drug prices. While manufacturing and procurement contribute to costs, distribution-related expenses account for about 60% of the final price to patients. In extreme cases, distribution costs account for 90% of a product's cost to patients.

Figure G3: Mark-ups for health products along the distribution chain



Note. Copied from [Silverman et al. \(2019\)](#)

Appendix H. Overview of the main medicine supply chain failures in LMICs

The complexity of health product supply chains in LMICs creates numerous opportunities for system failure. In this section, we adapt [Yadav \(2015\)](#) to categorize such challenges into distinct patterns that affect public and private sector medicine distribution differently.

Challenges in public sector medicine supply chains

Diffuse accountability represents perhaps the most fundamental weakness in public medicine distribution systems. When responsibility for supply chain performance is fragmented across multiple actors and administrative levels, each stakeholder can attribute failures to others in the system, creating a cycle where no single entity takes ownership of outcomes. Oversight mechanisms become ineffective when responsibility is dispersed. This accountability gap, combined with difficulties in tracking medication flows, can enable corruption in procurement and distribution, as well as diversion into private markets.

Financing uncertainty creates cascading disruptions throughout the procurement cycle. The timing of funding from ministries of finance, treasuries, or international donor agencies varies unpredictably, causing delays in the already lengthy procurement process. These financing delays can create system-wide stockouts as the entire supply chain struggles to synchronize with irregular funding flows. The resulting uncertainty makes it difficult for supply chain managers to plan effectively or maintain consistent service levels.

Structural complexity amplifies, rather than mitigates, supply chain vulnerabilities. Multiple tiers of stock holding and allocation decision-making worsen existing accountability gaps by adding more points where responsibility can be diffused. This multi-tiered approach also contributes to the "bullwhip effect," where small variations in patient demand at health clinics are amplified as order information flows upstream through the distribution system. The result is large swings in quantities produced and shipped at each stage, making it difficult for all tiers to remain synchronized with actual demand and ultimately requiring higher inventory levels to prevent stockouts.

Operational and political constraints further undermine system performance. Many procurement departments operate on annual cycles with limited flexibility in contracting mechanisms, forcing them to rely on imprecise forecasts that lead to stockouts or product expiration. While more frequent resupply could improve responsiveness, the associated transportation costs create trade-offs that are difficult to optimize without better demand forecasting. The chronic underfunding of operating expenses (including fuel, vehicle maintenance, and other logistical costs) reflects the political reality that these expenditures are less visible than capital investments, even though they are essential for system functionality.

Information gaps prevent effective supply chain management. Few public systems have processes to systematically capture consumption data, leaving supply chain planning based on outdated assumptions rather than real-time feedback. This lack of performance measurement creates weak incentives for supply chain staff, who operate without clear, measurable goals or accountability mechanisms. The mismatch between system design and local skills and capacity further complicates these challenges, as assumptions about the superiority of "pull systems" (where clinic staff place orders) often ignore the reality that frontline workers may lack the planning capacity to make these decisions effectively.

Challenges in private sector medicine supply chains

Geographic and economic inequities limit private sector reach in underserved areas. The small number of registered retail pharmacies in rural areas reduces government confidence in private sector distribution as a viable option for serving poor populations. Private distributors are often viewed as engaging in "cream skimming," focusing on high-value, profitable customers while leaving underserved populations to rely on inadequate public systems. While research suggests that private supply chains can reach remote areas when incentives are properly structured, current market dynamics often fail to achieve this potential.

Pricing structures create affordability barriers that may compromise treatment quality. Private sector prices frequently exceed international reference prices, making essential medicines unaffordable for large portions of the population. These high prices result from the presence of too many intermediaries in the supply chain and limited retail competition. While studies show that regulating retail and wholesale mark-ups can reduce prices without necessarily affecting drug availability, such regulatory interventions are often difficult to implement and enforce effectively.

Quality control and product selection reflect market rather than health system priorities. Weak regulation and poor enforcement lead to quality problems in private sector medicine distribution. Retail pharmacies, drug shops, and wholesalers often prioritize fast-moving products over essential medicines, creating gaps in the availability of treatments needed for public health priorities. The shortage of trained pharmacists exacerbates these problems, as many prefer to work in more lucrative areas of the supply chain, leaving rural areas dependent on informal drug sellers who may offer poor-quality products at high prices.

All of the above systematic failures in both public and private medicine supply chains can create significant barriers to effective healthcare delivery in the contexts we have studied. The interconnected nature of these challenges suggests that successful interventions require comprehensive approaches that address governance, financing, information systems, and market structures simultaneously.

Appendix I. Rough cost-effectiveness of Sierra Leone last-mile delivery

We reviewed the Sierra Leone last-mile vaccine delivery intervention in more detail after noting its apparently high effectiveness, which was rigorously evaluated through an RCT ([Meriggi et al., 2024](#)). The intervention aimed to increase COVID-19 vaccination rates in remote communities by deploying mobile vaccination teams and conducting community mobilization efforts (including local leader engagement and targeted outreach). The key findings of the RCT were:

- The intervention tripled the vaccination rate within 48-72 hours, increasing immunization by 26 percentage points in treated villages.⁴⁶
- The total cost per vaccinated person was \$33,⁴⁷ driven largely by transportation and personnel costs. The study also suggests that scaling this intervention would significantly reduce costs to \$23 per vaccinated person, as fixed costs would be spread across a larger population.
- The authors suggest that this intervention could be made more cost-effective by bundling it with other health services, such as HPV vaccination for girls aged 10–12 years and routine childhood immunizations (DTP, measles, polio), allowing for shared outreach and logistics costs, thereby reducing the per-dose delivery expense. However, they do not provide a specific estimate of the cost-effectiveness gains from this bundling. Note that research on this is currently being conducted ([SSRC, 2023](#)).

We developed a [rough model](#) to assess this intervention's cost-effectiveness in cG terms. We found that **it is likely substantially below cG's bar at ~400x**. This finding is driven by the estimated number of COVID-19 vaccine doses required to save one life in Africa, which is ~350 doses according to the best estimates we found ([Watson et al., 2022](#)).

Is it possible that this intervention surpasses cG's bar if it is bundled with other vaccines? We think this is unlikely to be the case. This would require a vaccine (or bundle of vaccines) that can save a life per no more than [70](#) administered doses. We extended our model by assuming that an HPV vaccine dose is administered together with a COVID-19 vaccine dose at no extra cost. Even with co-delivery, we estimate that 200 vaccine doses ([100](#) people receiving both vaccines) would be needed to save one life— still well above cG's 70-dose maximum threshold. In any case, we think a co-delivery scenario such as this is also overly optimistic, as we would expect the cost to increase at least slightly if several vaccines were bundled together.

Would this intervention be more cost-effective if we assumed that vaccines could be delivered by drones, thereby reducing transportation and personnel costs? Likely yes, but we still doubt that it could meet cG's bar. Haidari and co-authors estimated the cost savings that would come from using unmanned drones rather than land-based transportation methods to transport vaccines ([Haidari et al. 2016](#)). They found that, in Mozambique, drones would lead to roughly 20% savings per vaccine dose administered ([Bloomberg School, 2016](#)). Plugging this into our model, we found that a 20% reduction in costs would only increase the cost-effectiveness of the intervention to ~500x, which is still far from cG's bar.

⁴⁶ The intervention did not just work in the targeted villages, but many people from nearby villages also came to get vaccinated.

⁴⁷ Note that this does not include the costs of the vaccine doses, as these were provided by the COVAX program for free.

Appendix J. Detailed funder and implementer case studies

Funder case studies: PEPFAR and Gavi both increased their HSS focus in the mid-2000s due to political and donor pressure, with inconsistent success

This section examines how two major funders, [PEPFAR](#) and [Gavi, the Vaccine Alliance](#), navigated the transition to increased support for HSS. Both began as archetypal vertical programs, focused respectively on HIV/AIDS treatment and vaccine distribution, but in both cases the organization's leadership eventually felt that weak health system capacity was a major barrier to achieving their goals. Through a combination of internal evaluation, external criticism, and leadership-driven strategic shifts, these organizations incorporated HSS into their frameworks, though in different ways and with varying degrees of success. To analyze these transitions, we conducted a review of the available literature, including evaluations, policy analyses, and peer-reviewed studies. We examined how each funder approached HSS, what reforms were implemented, and what the outcomes have been, both in terms of health impact and system sustainability, to the extent that we could find information. We attempted to consider the political and financial pressures that influenced these shifts, as well as the operational challenges encountered in balancing vertical and systems-based approaches.

Our analysis of PEPFAR and Gavi's experiences with HSS revealed several recurring themes that highlight both the promise and the limitations of HSS as a donor strategy. In both cases, the transition **partially stemmed from financial sustainability concerns rather than an intrinsic commitment to systems reform**. In addition, the organizations' political and donor leadership strongly shaped the extent of HSS integration. In both cases, the **success of HSS efforts was uneven**, with integration easier in well-resourced settings and far more challenging in low-capacity environments. While HSS programs appear to have yielded some benefits, we **didn't find strong evidence that they improved health outcomes** (in line with the constant issues around HSS measurement) or that they improved long-term program sustainability.

PEPFAR

PEPFAR initially prioritized rapid, vertical scale-up of HIV/AIDS treatment and prevention, focusing on antiretroviral therapy (ART) delivery, prevention of mother-to-child transmission (PMTCT), and widespread HIV testing ([Bendavid, 2016](#), p. 3). This emergency-response model successfully expanded treatment access, with over 4 million people on ART and more than 40 million HIV tests conducted by 2011 ([Goosby et al., 2012](#), p. S54). However, **this approach relied heavily on U.S.-funded NGOs, creating parallel health systems that bypassed national healthcare infrastructure** in places like Tanzania ([Marten, 2015](#), abstract). Some experts were quite critical of PEPFAR's vertical structures, with complaints including logistical issues and reports that the program exacerbated critical health workforce shortages, with position vacancy rates exceeding 65% in HIV-affected regions ([Palen et al., 2012](#), p. S115).

By the late 2000s, PEPFAR and other global health donors began shifting resources from disease-specific programs to broader HSS ([Nattrass et al., 2016](#), p. 682). Sources claim that PEPFAR's **2008 congressional reauthorization explicitly emphasized HSS and workforce development**, aiming for deeper integration of HIV/AIDS services into national health systems ([USAID, 2021](#); [National Academy of Sciences, 2013](#)). Indeed, the reauthorization formally identified HSS as a priority, authorizing support for workforce training, capacity building, laboratory development, supply chain management, and public financial management systems

([Lantos and Hyde, 2008](#)).⁴⁸ The shift was driven by financial and logistical sustainability concerns, as donors began to feel that the infeasibility of maintaining donor-led vertical programs indefinitely was a serious concern ([Vogus & Graff, 2015](#), p. 276).

Eric Goosby, appointed as the United States Global AIDS Coordinator in 2009, seems to have played a strong role in reorienting PEPFAR towards a more integrated approach. On World AIDS Day 2009, Goosby announced a new five-year strategy designed to move PEPFAR toward “durable health systems” that could serve not only people with HIV/AIDS but also broader community health needs. This shift sought to engage national governments in capacity-building efforts to ensure that HIV/AIDS programs were not just externally funded and managed but integrated into national health systems ([News Medical, 2010](#)).

PEPFAR adapted by **shifting from direct service provision to strengthening governance, supply chains, and human resources for health**. To manage the transition, PEPFAR introduced Partnership Frameworks (PFs) in 21 countries, aligning its investments with national priorities, local governance, and healthcare infrastructures ([Palen et al., 2012](#), p. S114).⁴⁹

PEPFAR’s shift produced mixed but largely positive outcomes. In well-resourced countries like Botswana and South Africa, where national health infrastructure was relatively strong, **HIV care was successfully integrated into primary healthcare, allowing for sustainable, locally managed treatment and prevention services** ([Vermund et al., 2012](#), p. 2). However, in contexts such as Zambia, Mozambique, and Tanzania, **weak healthcare systems and workforce shortages made full transition to local ownership more difficult**, threatening continuity of care ([Vermund et al., 2012](#), p. 2; [Marten, 2015](#), abstract). In some cases, service integration led to inefficiencies: South African clinics that had previously relied on dedicated HIV services experienced increased waiting times and diminished quality of specialized adolescent HIV care after merging with general primary care ([Nattrass et al., 2016](#), p. 685). Another challenge of the transition was evaluating impact. As HIV services became integrated into national systems, it became harder to isolate their direct effects on health outcomes ([Bärnighausen et al., 2012](#)).

Despite these challenges, **PEPFAR’s investments in HSS yielded significant benefits**. By 2024, the program directly supported over 342,000 healthcare workers, bolstering both HIV service delivery and broader healthcare system resilience ([HIV.GOV, 2024](#)). The adoption of Sector-Wide Approaches (SWAs) in many countries demonstrated how donor-funded HIV interventions could be successfully embedded within national health systems, improving procurement, workforce management, and service delivery across multiple disease areas, as demonstrated in Malawi ([Palen et al., 2012](#), p. S115). Furthermore, PEPFAR-funded laboratory systems originally focused on HIV diagnostics expanded to include testing for tuberculosis, malaria, and sexually transmitted infections, improving overall diagnostic capacity and patient outcomes ([Palen et al., 2012](#), p. S117). PEPFAR has continued to explicitly include a systems emphasis, with the 2022-2023 strategy including “Public Health Systems and Security” as a central pillar ([PEPFAR, 2022](#), p. 17).

⁴⁸ Despite some sweeping descriptions of this formal policy shift, our impression of the legislative language of the reauthorization was that it was still quite vague on specific HSS commitments ([Lantos & Hyde, 2008](#)). While it authorized technical assistance for public finance management (Sec. 204), cooperative research and healthcare services (Sec. 205), and strengthening vaccine introduction protocols, clinical trials, and supply chains (Sec. 206), these provisions lacked explicit mechanisms to ensure deep structural reforms in national health systems.

⁴⁹ Despite the shift, much of PEPFAR’s funding and implementation remained focused on disease-specific interventions, rather than broader health systems transformation ([National Academy of Sciences, 2013](#)).

During its early years (2000–2005), Gavi’s investments led to rapid increases in immunization coverage, but these gains were often achieved through vertical donor-led mechanisms that bypassed national health policies and, in the eyes of some, weakened long-term resilience ([Sridhar & Tamashiro, 2009](#), p. 38). **Gavi’s partial transition from a strictly vertical funding model to a broader systems approach emerged from mounting evidence that systemic weaknesses in national health infrastructures were hindering vaccine delivery and long-term immunization sustainability.** As supply chain failures, workforce shortages, and governance inefficiencies became major bottlenecks, Gavi began to reassess its strategy ([Prosser et al., 2022](#), p. 2).

The push for systemic funding within Gavi appears to have been led by Julian Lob-Levitt, Gavi’s CEO from 2004 to 2010, who had extensive experience working with low-income country governments on health system reforms ([Wikipedia, 2024b](#)). He and a small group of colleagues advocated for HSS as a necessary condition for sustaining high vaccination rates, arguing that repeated, labor-intensive vaccination campaigns were unsustainable and disruptive to national health systems ([Storeng, 2014](#), p. 868). His arguments were reinforced by multiple evaluations of Gavi’s immunization programs, which revealed that vaccine deployment was often hampered by weak infrastructure, lack of trained personnel, unreliable transport, and poor data tracking on immunization coverage and vaccine stock levels ([Naimoli, 2009](#); referenced in [Storeng, 2014](#), p. 868).

Gavi’s board was deeply divided over whether HSS fit within its mandate.⁵⁰ In 2005, after an intense debate, Gavi narrowly voted to incorporate HSS into its funding strategy, marking a significant but contested policy shift ([Storeng, 2014](#), p. 869). The decision allocated up to 25% of Gavi’s budget for HSS, though **in practice, only around 10% of funding was actually spent on HSS initiatives** ([Tsai et al., 2016](#), p. 247). In 2010, Gavi also joined the Health Systems Funding Platform (HSFP), a multi-donor initiative with the Global Fund and Gates Foundation, intended to coordinate health system investments across global health actors ([Gavi, 2010](#)). However, Gavi’s funding for HSS within the HSFP remained tightly controlled, with funds still allocated primarily to immunization-focused infrastructure rather than broader health system development ([Storeng, 2014](#), p. 870).

Despite its shifts, **Gavi’s approach to HSS remained fragmented.** Some critics argued that HSS support diluted Gavi’s vaccine mission and questioned whether its governance structure was strong enough to ensure effective integration ([Hill et al., 2011](#), p. 8). Unlike the WHO’s six-building-block HSS framework, Gavi’s version of HSS prioritized measurable short-term inputs such as cold chain equipment and medical supplies, rather than broad structural reforms ([Tsai et al., 2016](#), pp. 250–251). This narrower definition of HSS reflected continued resistance from stakeholders who saw system-wide reform as outside Gavi’s mandate.

Gavi’s transition to an HSS model brought **some tangible improvements** in healthcare system integration and immunization sustainability. In countries like Sudan, the shift allowed for better integration of routine immunizations into the national healthcare system, improving overall vaccine accessibility and long-term sustainability ([Osman, 2021](#), p. 22). Gavi’s increased

⁵⁰ For example, Norway and the UK, both major donors, supported HSS as part of their broader commitments to system-wide health investments. In contrast, USAID and the Gates Foundation opposed the shift, arguing that HSS was an ill-defined, distracting concept that would dilute Gavi’s efficiency and slow progress on vaccine coverage ([Storeng, 2014](#), p. 868). Some critics within Gavi dismissed HSS as a public relations move to appease pro-HSS donors rather than a genuine shift in strategy ([Wikipedia, 2024b](#)).

engagement with national governments helped to **strengthen coordination** between ministries of health and immunization programs, **making vaccine delivery more resilient and reducing reliance on external partners** ([Galichet et al., 2010](#), pp. 214-215). Gavi's funding contributed to significant increases in diphtheria, pertussis, and tetanus (DPT) and measles (MMR) vaccination rates during this period, leading to reductions in infant and under-five mortality in recipient countries, although it is difficult to disentangle the effects of Gavi's HSS support from its vertical programs ([Jaupart et al., 2019](#), p. 7). These gains reinforced the argument that sustained immunization progress depends on robust healthcare infrastructure rather than one-off vaccination campaigns ([Naimoli, 2009](#), p. 5).

However, Gavi's transition to HSS has been criticized from many angles. In lower-capacity environments such as Chad and Cameroon, **evaluations found that weak financial management, disbursement delays, and governance inefficiencies severely undermined the effectiveness of HSS funding** ([Dansereau et al., 2017](#), pp. 6-8). In Cameroon, Gavi attempted to mitigate financial mismanagement by channeling funds through a third-party health partner rather than the Ministry of Health, but this created new tensions between stakeholders, increased administrative costs, and slowed down implementation ([El Bcheraoui et al., 2018](#), p. 10). Another key limitation of Gavi's HSS approach was that, in practice, **funding continued to prioritize short-term immunization-related spending** rather than deeper structural reforms. A 2016 analysis of Gavi's grants found that more than half of HSS funds were allocated to purchasing medical supplies, equipment, and facilities, with very little funding dedicated to operational research, efficiency improvements, or national policy development ([Tsai et al., 2016](#), pp. 250-251). Despite efforts to integrate immunization with broader health system development, Gavi and other global health partnerships struggled to ensure that vaccines reached marginalized populations equitably, raising concerns about persistent inequities in immunization coverage ([Nunes et al., 2024](#), p. 10).

Gavi's HSS support also did not seem to meaningfully improve program sustainability. Critics argued that, instead of transforming health systems, HSS spending was often used to support short-term immunization efforts, making long-term sustainability difficult ([Mimche et al., 2017](#), p. 411). Many governments have struggled to maintain immunization programs after Gavi support phased out ([Kallenberg et al., 2016](#), abstract). Kenya, for example, faced major financing and policy challenges in maintaining its pneumococcal vaccination program once Gavi's direct funding ended, demonstrating how difficult it was for countries to transition to independent immunization financing ([Ojal et al., 2019](#), p. e653).

Four implementer case studies reveal mixed potential for vertical programs to benefit from systems integration

This section aims to understand the potential for health systems interventions to benefit vertical programs, as well as the potential upsides of integrating vertical programs into larger health systems. We spent two hours searching various combinations of relevant search terms on Google, Google Scholar, Elicit, and ChatGPT to generate a list of potential case studies, and selected four case studies representing a diversity of vertical health interventions, varying levels of governmental support, and a variety of perceived levels of success.⁵¹

⁵¹ We came across several works by Results for Development (R4D) that summarize additional case studies: Integrating Family Planning into Public Healthcare (PHC; [Rowan et al., 2019](#)) and Integrating vertical programs into PHC ([R4D, 2019](#)) (Bangladesh, India, South Africa, Ethiopia). We have focused on different case studies for our exercise, but recommend the R4D reports for interested readers.

The Onchocerciasis Control Programme (OCP) was launched in 1974 as a collaboration between WHO, UNDP, the World Bank, and FAO to combat river blindness in West Africa. It relied on aerial spraying of insecticides to eliminate blackfly larvae, adding ivermectin (Mectizan®) treatment in 1987, which ultimately became the program's primary intervention. OCP halted disease transmission in 10 of 11 countries but operated vertically, with minimal integration into national health systems ([WHO, 2007](#); [Msuya, 2003](#), p. 9).

In 1995, the WHO and partners folded the OCP into the new African Programme for Onchocerciasis Control (APOC), which expanded control efforts to 30 additional countries and introduced Community-Directed Treatment with Ivermectin (CDTI). In order to facilitate rapid scale-up, the **new model integrated onchocerciasis control into primary healthcare**, shifting responsibility for drug distribution to local workers (Community Drug Distributors, or CDDs, usually unpaid) **as part of the broader healthcare system** ([Coffeng et al., 2013](#), p. 8; [WHO, 2015](#), p. 25). Beyond onchocerciasis control, APOC's CDTI strategy **supported local healthcare systems**, benefiting other priority healthcare programs such as deworming, malaria control, maternal and child health services, and immunization ([Homeida et al., 2002](#), abstract).

The program's horizontal reorientation and scale-up were **widely viewed as a success**: annual treatment coverage expanded from 8 million (1996, at the start) to 20 million just four years later ([Seketeli, 2002](#), p. 4). By 2015, the program had delivered over 930 million ivermectin treatments, reducing onchocerciasis prevalence from 45% to 18% and blindness prevalence from 0.6% to 0.2% ([Coffeng et al., 2013](#), p. 4). These reductions translated into an estimated 17.4 million DALYs averted between 1995 and 2015, at a cost of approximately \$478 million, or an average cost of \$27/DALY (2012 international dollars, [Coffeng et al., 2013](#), Table 2). The indirect effects of mass ivermectin drug administration on parasitic infections, school attendance, and economic productivity further reinforced the perception of APOC's wide-ranging public health benefits ([Amazigo, 2008](#), abstract; [Krotneva et al., 2015](#), p. 8).

APOC's CDTI structure still faced challenges, including CDD volunteer fatigue ([WHO, 2007](#), p. 23), as well as competition for resources due to the HIV/AIDS crisis ([WHO, 2007](#), p. 29; [Msuya, 2003](#), p. 15). By early 2016, onchocerciasis elimination remained out of reach in eight regions covering 10.4 million people (out of the region's 86 million), requiring treatment extensions for these regions ([Tekle et al., 2016](#), p. 1; [Noma et al., 2014](#), p. 10).

In late 2015, at the end of its 20-year mandate, APOC was replaced by the Expanded Special Project for Elimination of Neglected Tropical Disease ([ESPEN](#)), which expanded beyond onchocerciasis to other neglected tropical diseases (NTDs), including lymphatic filariasis and trachoma ([WHO, 2016](#)). During its five-year mandate, ESPEN focused on health system strengthening, including training health workers, improving supply chains, and strengthening surveillance ([WHO, 2021](#); [WHO, 2016](#)). We have not looked into the reasons why ESPEN was wound down at the end of its five-year period, but suspect that it was related to frequently-cited funding concerns. In place of ESPEN's work, 25 countries in Africa established National Onchocerciasis Elimination Committees (NOECs) to manage progress toward elimination at the national level, and the WHO maintains global elimination by 2030 as a goal, supported by the Global Onchocerciasis Network for Elimination (GONE) ([WHO, 2025a](#); [2025b](#)).

In the 1990s and 2000s, Malawi's healthcare system included many donor-driven vertical programs ([Sakala et al., 2022](#), p. 1). The programs and implementers were different for each disease, including tuberculosis⁵² and HIV.⁵³ In 2011, the Malawi government introduced the National Health Sector Strategic Plan (HSSP) for 2011–2016, emphasizing the need for a comprehensive approach to healthcare ([Global Financing Facility, 2022](#), p. 53).

Malawi's integration of HIV, family planning, maternal and child health, and sexually transmitted infection (STI) services was motivated by **clear inefficiencies in vertical programming**. For example, evidence suggested that integrating HIV/AIDS services with maternal health and family planning would improve service uptake, including higher antiretroviral therapy (ART) initiation ([Pfeiffer et al., 2010](#), p. 4), increased contraceptive use ([Lindegren et al., 2012](#), p. 17), and reduced perinatal HIV transmission ([Car et al., 2011](#), p. 8). One serious weakness was in pediatric HIV follow-up, where only 29.5% of HIV-exposed infants were enrolled in care, and 34.2% were lost to follow-up or died, showcasing the limited success of early integration efforts in preventing perinatal transmission ([Braun et al., 2011](#), p. 5).

The **country integrated vertical HIV, tuberculosis, and malaria programs into broader reproductive, maternal, newborn, child and adolescent health (RMNCAH) services**, aiming to align disease-specific initiatives with primary healthcare ([Global Financing Facility, 2022](#), p. 53). Of the pre-integration vertical and programmatic implementers we found, all appear to still be involved post-integration. The most recent HSSP II 2017–2022 explicitly focused on integrating vertical programs, such as HIV/AIDS, tuberculosis, malaria, and immunization services, into the Essential Health Package (EHP) to improve reproductive, maternal, newborn, child, and adolescent health (RMNCAH) outcomes ([Global Financing Facility, 2022](#), p. 14).

The integration appears to have had **mild but positive results so far**, although we have not found high-quality evidence that the program has been a success. ART enrollment more than doubled in integrated antenatal care settings ([Suthar et al., 2013](#), p. 48), and family planning uptake rose by 14%, although immunization rates remained mostly unchanged ([Cooper et al., 2020](#)). While integration does appear to have improved access to a variety of services, and uptake for many, further health system bottlenecks (such as patient follow-up failures, low patient retention for infants, and long lines at clinics) have prevented major reductions in maternal and infant mortality ([Suthar et al., 2013](#), pp. 46, 51).

Africa AHEAD and Community Health Clubs

Africa AHEAD (Applied Health Education and Development) is an NGO dedicated to improving public health in SSA through community-based initiatives ([Africa AHEAD](#)). Originally, the organization focused narrowly on water and sanitation interventions. However, recognizing that hygiene behavior change involved a range of interconnected factors and required sustained community and public health engagement, Africa AHEAD **transitioned from a vertical to a horizontal approach, supporting governments to incorporate water, sanitation, and hygiene (WASH) efforts/initiatives/interventions into integrated health strategies** ([Africa AHEAD](#)).

⁵² These include the National Tuberculosis Control Programme (NTCP) ([Simwaka et al., 2007](#)), USAID and Project HOPE ([Sakala et al., 2022](#)), [FHI 360 \(2022\)](#), and Partners In Health ([Wikipedia, 2025c](#)), most of which stressed that they have always closely coordinated with the Malawian government.

⁵³ These include [FHI 360 \(2022\)](#), [Jhpiego \(2019\)](#), mothers2mothers ([Wikipedia, 2024c](#)), and the Community of Sant'Egidio (DREAM Program; [Wikipedia, 2025d](#)), most of which stressed that they have always closely coordinated with the Malawian government.

In 1995, the first Community Health Clubs (CHCs, Africa AHEAD's horizontal model) were established in Zimbabwe and were initially designed as grassroots organizations to support CHWs and other primary healthcare staff with local community leadership ([Wikipedia, 2023c](#)). Although initially intended to focus just on WASH, the CHC model now involves a holistic approach to health, and several national governments have adopted the model as a key component of public health strategy. By 2015, over 3,000 CHCs had been established, involving more than 2 million people in Zimbabwe alone ([Wikipedia, 2023c](#)).

In 2009, Rwanda's Ministry of Health launched the Community-Based Environmental Health Promotion Programme (CBEHPP), aiming to establish CHCs in all 15,000 villages nationwide. By 2014, over 90% of these villages had registered CHCs, with more than 5,000 communities receiving training. The program's goal was to benefit approximately 9 million people, about 80% of the population, by 2016 ([SuSanA, 2016](#)), although we have not yet found evaluations of whether this has been successful.

There's some evidence that the integrated CHC model has an impact on WASH-specific knowledge and behavior.⁵⁴ Catholic Relief Services claims that the clubs have been effective in disseminating vital health information and practices at the community level. ([Catholic Relief Services](#)). [Rosenfeld et al. \(2021\)](#), in a review, found that CHC members consistently adopted 9–16 more recommended WASH behaviors than control groups. In Zimbabwe, CHCs significantly increased latrine ownership and safe sanitation practices (93.4% vs. 43.2% in controls). Similarly, [Waterkeyn et al. \(2019\)](#) found that CHCs resulted in over 90% compliance with 12 key hygiene practices in Zimbabwe and 80% adoption of 10 new practices in Rwanda.

Beyond hygiene behavior, CHCs appear to be at least somewhat effective and have been **linked to reductions in diarrheal diseases and other communicable conditions**. [Ntakirutimana et al. \(2022\)](#) found significant reductions in diarrhea (82.8%), soil-transmitted helminths (74.2%), and malnutrition (96%) following CHC implementation in Rwanda. In Vietnam, [Waterkeyn et al. \(2021\)](#) found that diarrheal disease cases fell by 117 in CHC communities compared to just 24 in non-CHC communities. [Waterkeyn \(2004\)](#) found that, in Ruombwe, Zimbabwe, diarrhea cases declined from 404 to 38, and bilharzia was nearly eliminated (1,310 cases to just 1 case) after CHC implementation.

It appears that the CHC model may have lost momentum until recently. A 2024 article references that CHCs in Zimbabwe had been “revived” by the Ministry of Health and Child Care and international partners to combat a range of public health and economic challenges ([Mutsaka, 2024](#)). Additionally, it appears that Family Action for Community Empowerment in Zimbabwe (FACE Zim), in partnership with Africa AHEAD-Zimbabwe and the Ministry of Health, recently organized events to further community health initiatives ([Facebook, 2024](#)).

⁵⁴ It is hard to compare the effectiveness of CHCs to other WASH methods, and some evidence suggests that WASH in general may not improve health outcomes. For example, [Ramesh et al. \(2015\)](#) found strong short-term effects of emergency WASH programs in humanitarian settings, with interventions reducing diarrheal disease by 90% in some studies. [Whaley and Webster \(2011\)](#) found that CHCs were more effective at promoting handwashing, while CLTS was more effective at latrine construction. A meta-analysis found that WASH interventions (potentially including CHCs) had no significant effect on weight-for-age or weight-for-height in children, though they did find marginal gains in height-for-age ([Dangour et al., 2013](#)).

We spoke with Adam Salisbury at GiveWell, which has funded the Malaria Consortium to carry out seasonal malaria chemoprevention (SMC) campaigns in multiple countries, including Mozambique. These campaigns aim to distribute preventive malaria treatment to children during peak transmission seasons.

While some observers describe bednet distribution as operating in a more vertical model, Salisbury noted that both SMC and bednet campaigns typically **work quite closely with government systems**. In Mozambique, for example, an SMC campaign relied on CHWs, who are technically government employees but frequently support donor-funded programs. GiveWell's funding facilitated drug procurement and supply chain logistics managed by Malaria Consortium. It also supported training and mobilizing CHWs to conduct household visits and administer medication, while enabling collaboration with Mozambique's Ministry of Health to align malaria prevention efforts with national epidemiological data. While the program integrated CHWs into its operations, higher-level supply chain and distribution functions remained donor-managed, reducing reliance on government procurement systems.

Despite its success in delivering malaria treatment, the Mozambique SMC campaign faced **significant bottlenecks due to political instability**. According to Salisbury, the program was canceled, possibly “because of election violence.” This highlights a key risk: **even partially integrated vertical programs remain exposed to government instability and external disruptions**. Additionally, while Malaria Consortium's SMC programs leverage existing health systems, they do not necessarily strengthen them in a sustainable way. If GiveWell or Malaria Consortium exit, local health systems may not have the resources to continue the program independently.